Let’s power up reforming

WITH eREACT™ TECHNOLOGY

BAF briefing – March 27, 2026

TOPSOE

TOPSOE ON THE FOREFRONT OF ELECTRIFIED REFORMING DEVELOPMENT

Topsoe eREACT™ journey

- 2014: Development of eREACT™ initiated
- 2017: First patent filed
- 2019: First of a kind paper describing eSMR in use **published in *Science***[1]
- 2021: Start-up of **pilot plant** in Foulum
- 2021: First none-Topsoe electrified reforming technologies **announced**[2]
- 2022: Topsoe & Aramco announce collaboration to demonstrate eREACT™ and Pd membrane
- 2024: **Engineering package for 2000 BPD** eFuels completed
- 2024: Start-up of the first none-Topsoe eSMR and eCracker **pilot plants**[3]
- 2025: **DLR demo FID'ed and under construction**
- 2025: >50 patents filed
- 2025: Supply chain in place for global launch (commissioning of new catalyzed hardware facility)
- 2025: Announcement of first none-Topsoe **commercial projects**[4]

Global developments within electrified reforming

1: Electrified methane reforming: A compact approach to greener industrial hydrogen production | Science 2: To Topsoe's knowledge. Announcements in 2021 include Sypox being founded, BASF, SABIC and Linde signing a JDA for electrically heated steam cracker furnaces 3: To Topsoe's knowledge. Pilots include BASF, SABIC, and Linde's electrically heated steam cracking furnace and Sypox eSMR pilot; 4: See for example Sypox and European Energy and NEXTCHEM

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eREACT™ IS HIGH ON THE AGENDA FOR MISSION 2026

PROTECT PROFITABILITY OF TOPSOE *Main effort*	• **Reduce total cost of catalysts,** DKK 150m • **Cost / Lead time reduction of engineering packages (GC & PtX)**, 40% & 75% • **Reduce cost of WSA hardware**, 40% • **Productivity improvement**, 10% • **Increase Quality Maturity**, 3.0 • **Preserve Cash**, Free Cash Flow of at least DKK -500m
DRIVE UTILIZATION OF NEW ASSETS	• **Power-to-X accumulative backlog inflow,** DKK 1.2bn • **Reach high case budget volume for TK**, 26,500 ton • **Fill backlog of eREACT™ catalyzed hardware**, >3MW by end of year 2026
DELIVER SUSTAINABLE IMPROVEMENTS IN WORKPLACE SAFETY	• **25% reduction in recordable incidents vs 2025,** ≤17 • **90% closure of all hazard reports within 30 days**

Head of eREACT™

2026 Mission Summary

"Secure first commercial eREACT™ supply contracts to drive profitability through step-change in asset utilization"

PRIORITIES

Backlog on supply of catalyzed hardware (3 MW)

Paid Studies within Hydrogen, Methanol and Ammonia, including revamps (8 studies)

Paid Studies with eSAF (5 studies)

Secure eREACT™ competitiveness

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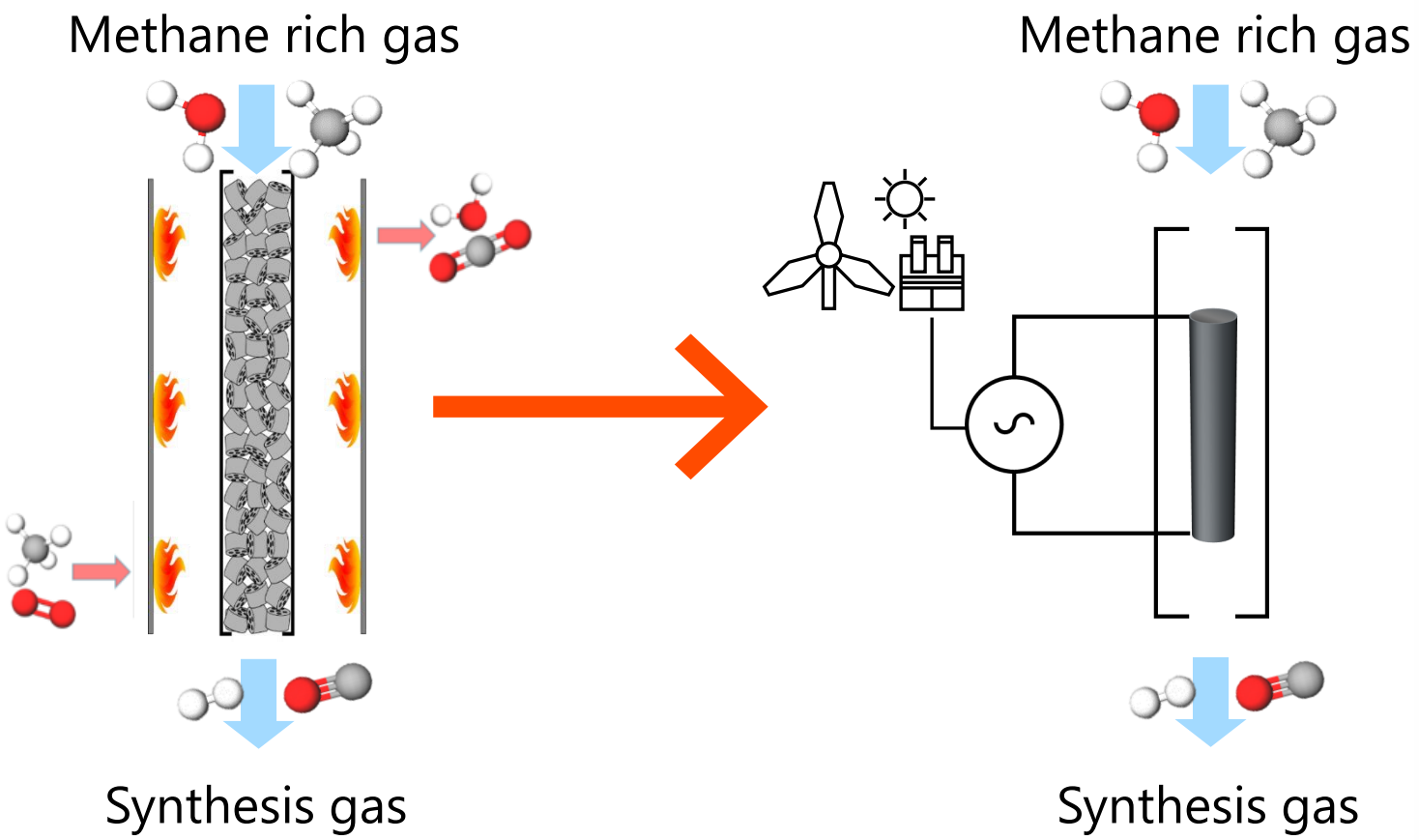
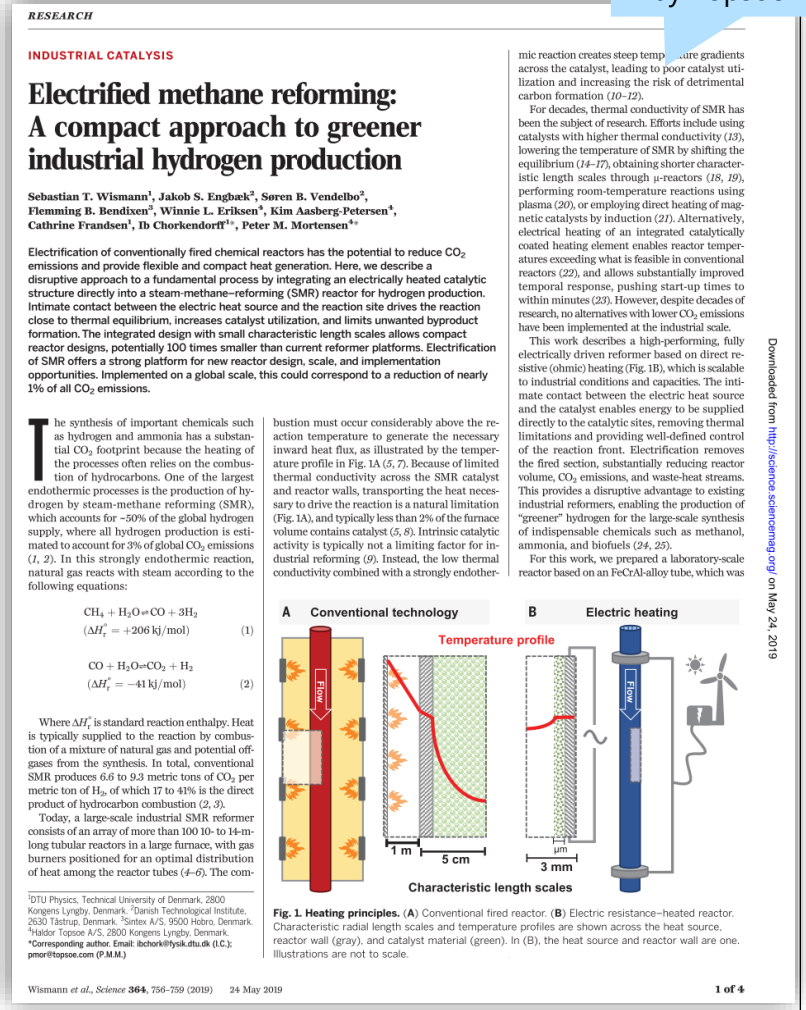
THE FRONT-RUNNER ELECTRIFIED REFORMING TECHNOLOGY | eREACT™ REMOVES THE FIRED SECTION OF CHEMICAL REACTORS THROUGH ELECTRIFICATION

First-of a kind paper in ***Science*** published by Topsoe in 2019





RESEARCH

INDUSTRIAL CATALYSIS

Electrified methane reforming: A compact approach to greener industrial hydrogen production

Sebastian T. Wismann[1], Jakob S. Engbæk[2], Søren B. Vendelbo[2], Flemming B. Bendixen[3], Winnie L. Eriksen[4], Kim Aasberg-Petersen[4], Cathrine Frandsen[1], Ib Chorkendorff[1]*, Peter M. Mortensen[1]*

Electrification of conventionally fired chemical reactors has the potential to reduce CO_2 emissions and provide flexible and compact heat generation. Here, we describe a disruptive approach to a fundamental process by integrating an electrically heated catalytic structure directly into a steam-methane–reforming (SMR) reactor for hydrogen production. Intimate contact between the electric heat source and the reaction site drives the reaction close to thermal equilibrium, increases catalyst utilization, and limits unwanted byproduct formation. The integrated design with small characteristic length scales allows compact reactor designs, potentially 100 times smaller than current reformer platforms. Electrification of SMR offers a strong platform for new reactor design, scale, and implementation opportunities. Implemented on a global scale, this could correspond to a reduction of nearly 1% of all CO_2 emissions.

The synthesis of important chemicals such as hydrogen and ammonia has a substantial CO_2 footprint because the heating of the processes often relies on the combustion of hydrocarbons. One of the largest endothermic processes is the production of hydrogen by steam-methane reforming (SMR), which accounts for ~50% of the global hydrogen supply, where all hydrogen production is estimated to account for 3% of global CO_2 emissions (*1*, *2*). In this strongly endothermic reaction, natural gas reacts with steam according to the following equations:

$$CH_4 + H_2O \rightleftharpoons CO + 3H_2 \quad (\Delta H_r^\circ = +206\ \mathrm{kJ/mol}) \quad (1)$$

$$CO + H_2O \rightleftharpoons CO_2 + H_2 \quad (\Delta H_r^\circ = -41\ \mathrm{kJ/mol}) \quad (2)$$

Where ΔH_r° is standard reaction enthalpy. Heat is typically supplied to the reaction by combustion of a mixture of natural gas and potential off-gases from the synthesis. In total, conventional SMR produces 6.6 to 9.3 metric tons of CO_2 per metric ton of H_2, of which 17 to 41% is the direct product of hydrocarbon combustion (*2*, *3*).

Today, a large-scale industrial SMR reformer consists of an array of more than 100 10- to 14-m-long tubular reactors in a large furnace, with gas burners positioned for an optimal distribution of heat among the reactor tubes (*4*–*6*). The combustion must occur considerably above the reaction temperature to generate the necessary inward heat flux, as illustrated by the temperature profile in Fig. 1A (*5*, *7*). Because of limited thermal conductivity across the SMR catalyst and reactor walls, transporting the heat necessary to drive the reaction is a natural limitation (Fig. 1A), and typically less than 2% of the furnace volume contains catalyst (*5*, *8*). Intrinsic catalytic activity is typically not a limiting factor for industrial reforming (*9*). Instead, the low thermal conductivity combined with a strongly endothermic reaction creates steep temperature gradients across the catalyst, leading to poor catalyst utilization and increasing the risk of detrimental carbon formation (*10*–*12*).

For decades, thermal conductivity of SMR has been the subject of research. Efforts include using catalysts with higher thermal conductivity (*13*), lowering the temperature of SMR by shifting the equilibrium (*14*–*17*), obtaining shorter characteristic length scales through μ-reactors (*18*, *19*), performing room-temperature reactions using plasma (*20*), or employing direct heating of magnetic catalysts by induction (*21*). Alternatively, electrical heating of an integrated catalytically coated heating element enables reactor temperatures exceeding what is feasible in conventional reactors (*22*), and allows substantially improved temporal response, pushing start-up times to within minutes (*23*). However, despite decades of research, no alternatives with lower CO_2 emissions have been implemented at the industrial scale.

This work describes a high-performing, fully electrically driven reformer based on direct resistive (ohmic) heating (Fig. 1B), which is scalable to industrial conditions and capacities. The intimate contact between the electric heat source and the catalyst enables energy to be supplied directly to the catalytic sites, removing thermal limitations and providing well-defined control of the reaction front. Electrification removes the fired section, substantially reducing reactor volume, CO_2 emissions, and waste-heat streams. This provides a disruptive advantage to existing industrial reformers, enabling the production of "greener" hydrogen for the large-scale synthesis of indispensable chemicals such as methanol, ammonia, and biofuels (*24*, *25*).

For this work, we prepared a laboratory-scale reactor based on an FeCrAl-alloy tube, which was

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Fig. 1. Heating principles. (**A**) Conventional fired reactor. (**B**) Electric resistance–heated reactor. Characteristic radial length scales and temperature profiles are shown across the heat source, reactor wall (gray), and catalyst material (green). In (B), the heat source and reactor wall are one. Illustrations are not to scale.

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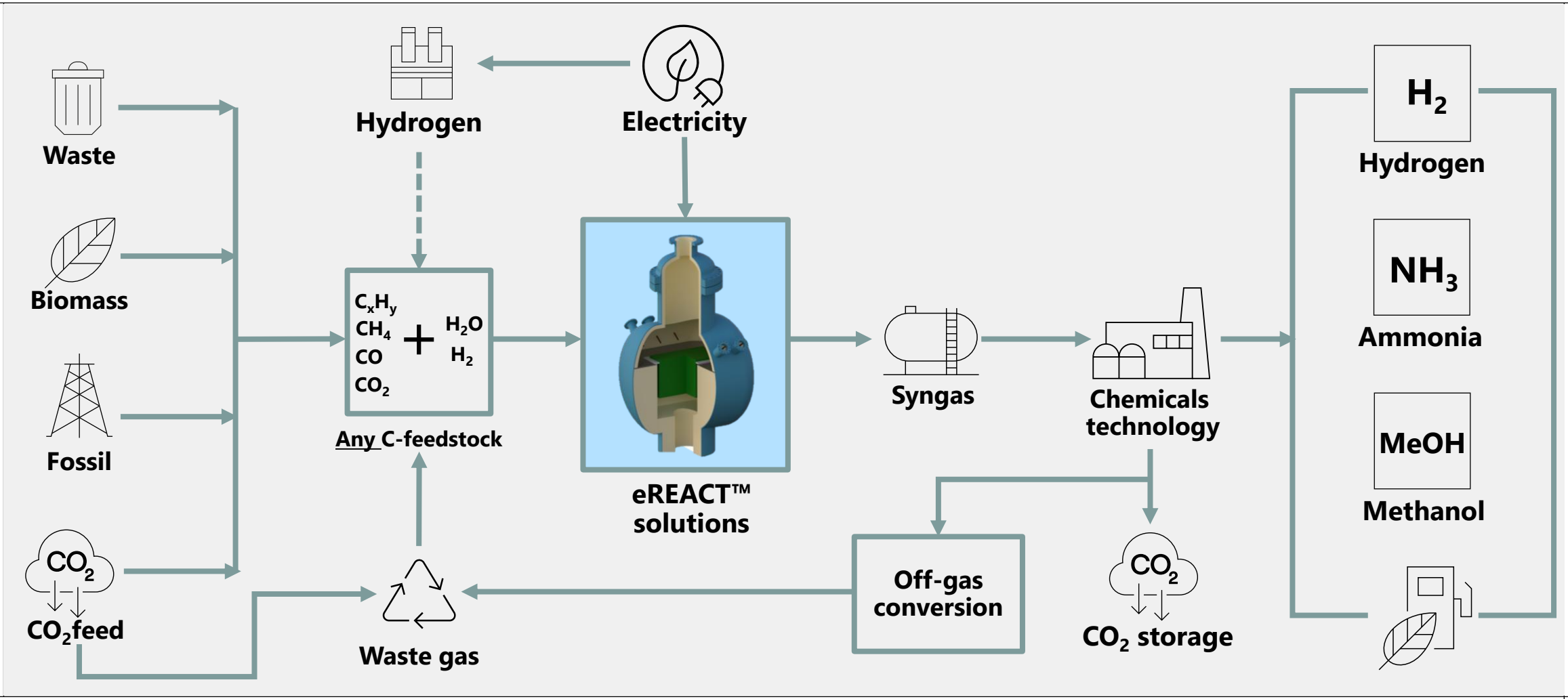
eREACT™ A PLATFORM TECHNOLOGY WITH MULTIPLE APPLICATIONS



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eREACT™ IS ADDING A TECHNOLOGY TO THE WIDEST PORTFOLIO OF REFORMING TECHNOLOGIES

SynCOR™
ADVANCED AUTOTHERMAL REFORMING
100,000 - 820,000 Nm³/h

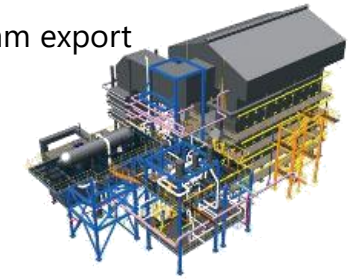
- 99%+ Carbon recovery
- Lowest LCOH

SMR
RADIANT WALL TUBULAR REFORMER
20,000-200,000 Nm³/h

- Steam export flexibility
- Medium range capacity

SMR-B
SMR WITH BAYONET TUBES
20,000-200,000 Nm³/h

- SMR with Bayonet tubes
- Low / zero steam export

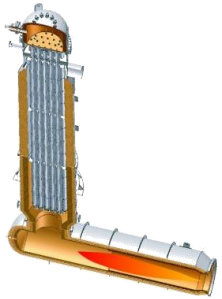


HTER
HALDOR TOPSOE EXCHANGE REFORMER
50,000- 250,000 Nm³/h

- Reduced steam export
- Option for plant revamp to increase capacity

ModuLite™
CONVECTION REFORMER
1,000-50,000 Nm³/h

- Zero steam export
- Compact plot



eREACT™
ELECTRIFIED REFORMER
3-100,000+ Nm³/h

- Zero flue gas emissions
- Rapid start-up

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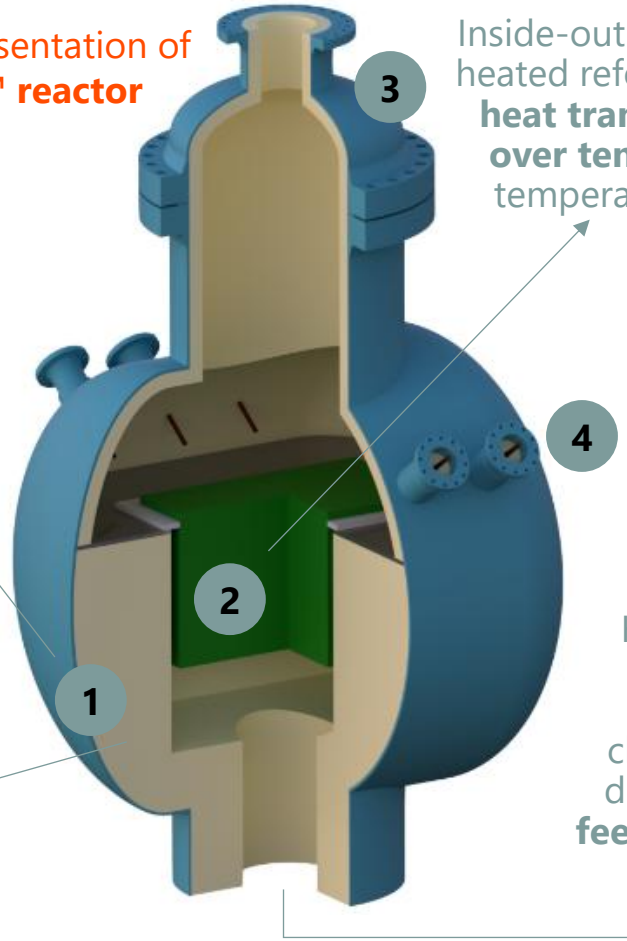
eREACT™ DESIGNED FOR OPTIMAL PERFORMANCE AND SAFETY

eREACT™ reactor basics

- Refractory-lined pressure shell (1)
- Feed gas enters the reactor from the top, and product gas exits at the bottom at the desired stoichiometry
- Catalyzed hardware (2) installed in the reactor via the removable top dome (3)
- Feedthroughs (4) provide the power from outside the reactor to the catalyzed hardware

Schematic representation of the **eREACT™ reactor**



Inside-out intimate electrically heated reforming for **efficient heat transfer** and **minimal over temperature** in high temperature environment

30 barg pressure vessel for integration in **known syngas process schemes** with low alloy steel

High single pass conversion and stochiometric chemicals process design with **>95% feedstock utilization**

Refractory lining provides **safe proven** method of housing 1000°C reaction environment and allows **99% energy efficiency**

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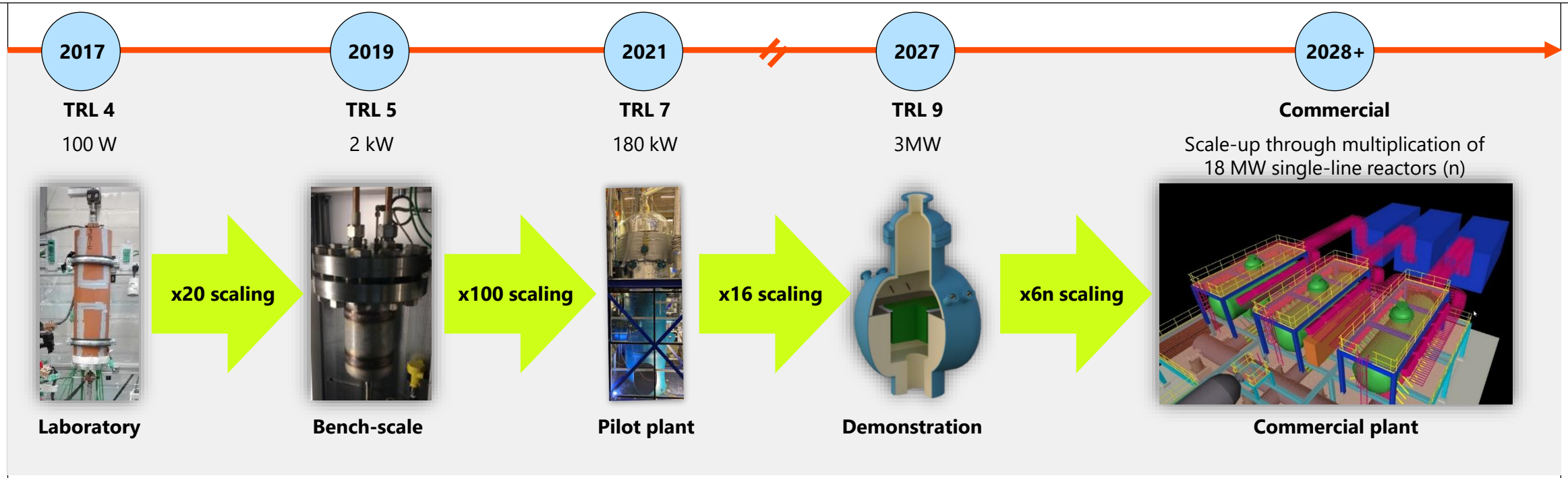
EXTENSIVELY TESTED TECHNOLOGY READY FOR COMMERCIAL DEPLOYMENT



Majority of eREACT™ technology constituents are already at TRL 9 and have been utilized in other commercial Topsoe technologies. Current overall TRL rating based on the eREACT™ novel catalytic system which is only constituent at TRL 7[1]

eREACT™ constituent	TRL rating	eREACT™ constituent	TRL rating	eREACT™ constituent	TRL rating	eREACT™ constituent	TRL rating
Catalyzed hardware	**TRL 7**	Refractory lining	TRL 9	Pressure vessel	TRL 9	Power supply unit (PSU)	TRL 9
Catalyst	TRL 9	Temperature control	TRL 9	Feed-throughs and terminal boxes	TRL 9		

1: Actual catalyst and carrier are well known in industrial operation

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eREACT™ PILOT PLANT IN FOULUM DENMARK HAS BEEN OPERATING SINCE AUGUST 2021

Extensive experiments conducted in the pilot plant to:

→ Verify all key elements in the novel reactor

→ Demonstrate the performance at varying operating conditions

→ Demonstrate catalyst stability

→ Test transients and upset situations including trips

→ Test feed variations

Pilot plant has run for >15,000 hours
(and counting)

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>15,000 HOURS OF PILOT PLANT TESTING HAS SHOWN THAT eREACT™ PERFORMS AS PREDICTED UNDER VARIOUS OPERATING PARAMETERS

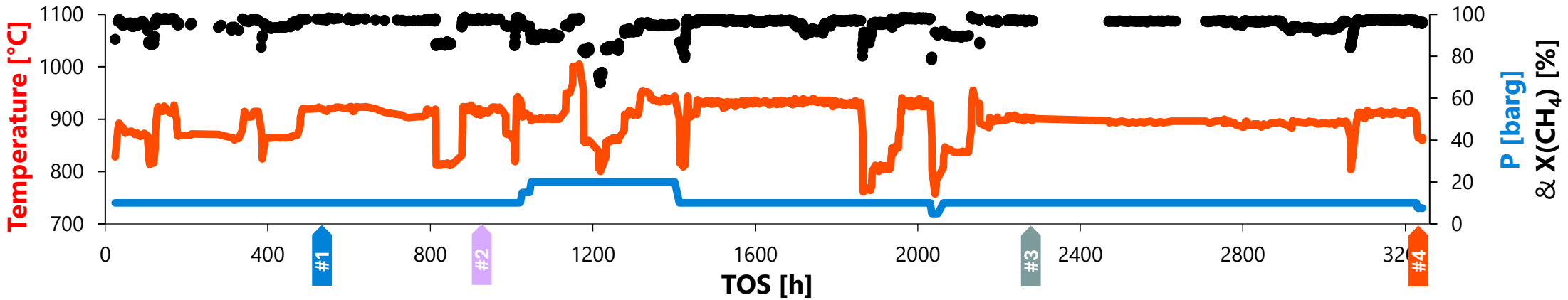

Figure showing almost 3500 hour run in eREACT™ pilot. T is temperature, P is pressure and X(CH_4) is methane conversion

	#1	#2	#3	#4
TOS [h]	521	946	2267	3211
Temperature [°C]	910	910	910	910
CH_4 conversion [%]	97.8 ± 0.1	97.7 ± 0.3	97.1 ± 0.1	97.7 ± 0.1

Table showing key performance data from almost 3500 hours run in eREACT™ pilot (same as above). Note that the biogas composition varies with time

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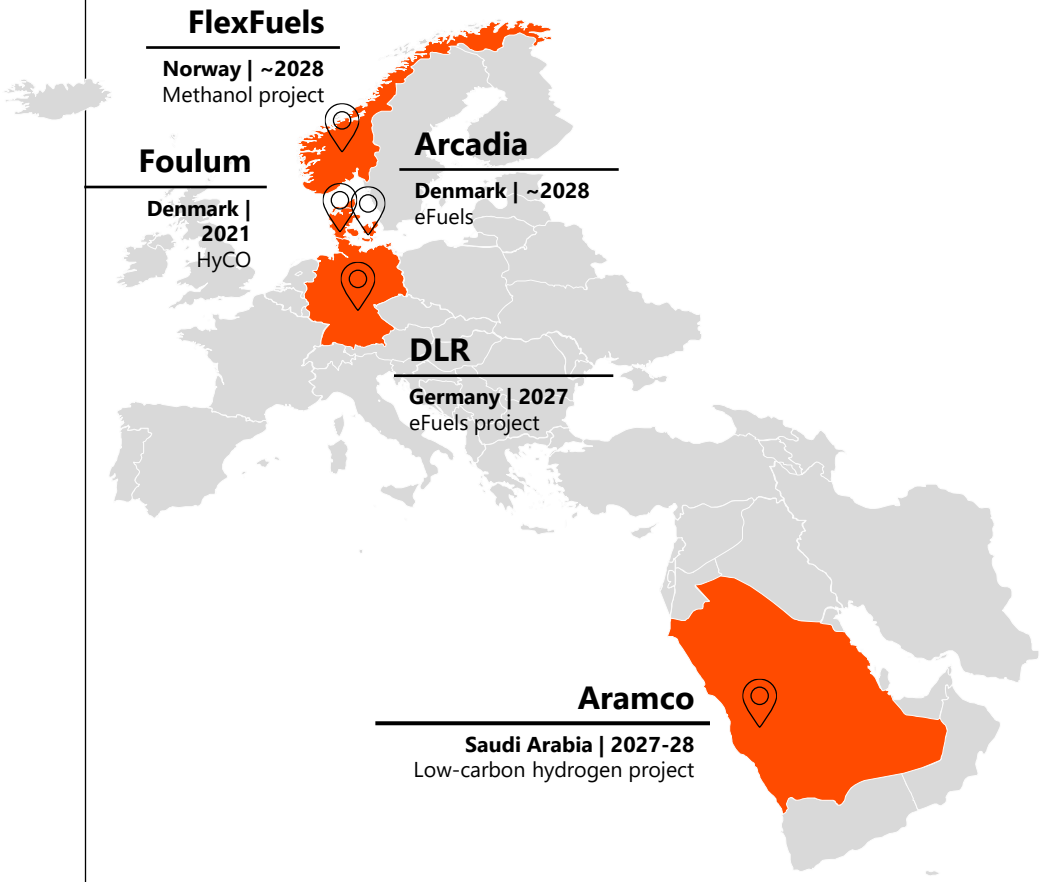
COMMERCIALIZATION OF eREACT™ SUPPORTED BY STRONG PORTFOLIO OF ACTIVE PROJECTS



Selected eREACT™ related projects

Owner	Scale	Product	Capacity	Stage	Expected start-up
Topsoe	Pilot	HyCO	0.18 MW	Operational	In operation from 2021
Aramco	Demo/JDA	Hydrogen	3 MW (3kNm3/h)	FEED complete	2027-28
DLR	Demo	eSAF (FT route)	3 MW (100 BPD)	Under construction	2027
FlexFuels	Commercial	Methanol	15 MW (150 MTPD)	FEED complete	Awaiting FID
Arcadia	Commercial	eFuels (FT route)	45 MW (2000 BDP)	FEED complete	Awaiting FID

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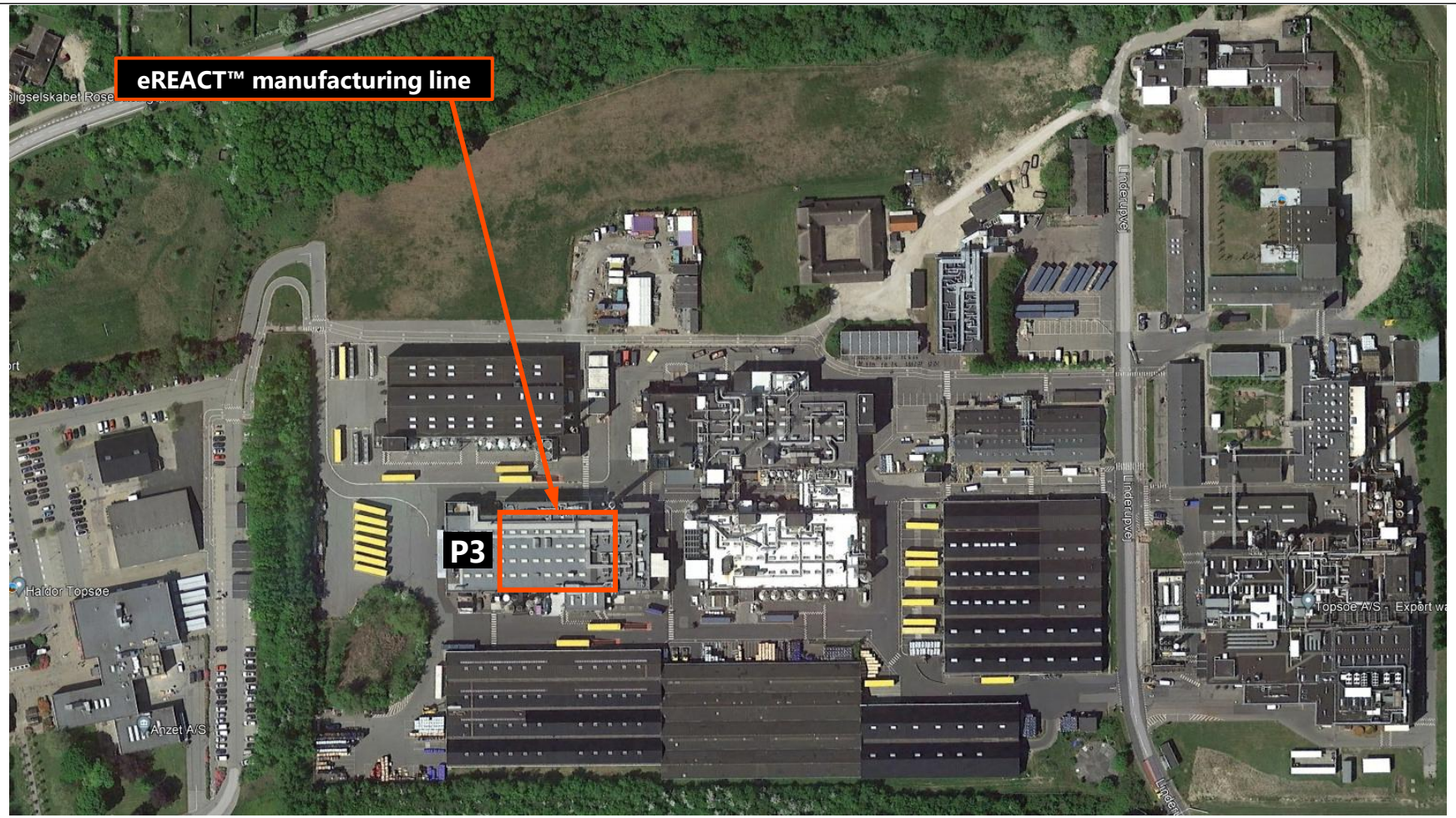
eREACT™ INDUSTRIAL MANUFACTURING LINE BUILT INTO TOPSOE'S FREDERIKSSUND CATALYST FACTORY



Picture of Topsoe's catalyst manufacturing plant in Frederikssund, Denmark

- Largely automated manufacturing process vastly relying on robots, allowing for industrial mass production
- Final product: eREACT™ Catalyzed Hardware
- Commissioned in Q1-2026

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ELECTRIFIED REFORMING COMPETITION IS INCREASING

In operation | In construction | Announced

COMPETITOR	STAGE: Pilot	STAGE: Demo	STAGE: Commercial	ANNOUNCED APPLICATIONS: Hydrogen	Biogas to syngas	eFuels	eCracking	Other
TOPSOE	180kW	3MW (2500 mtpa SAF)	15-45MW	☑	☑	☑	☐	☐
SYPOX	22kW	250kW	10MW	☑	☑	☐	☐	☐
NextChem			15 kNm^3/hr H_2	☑	☐	☐	☐	☐
Ineratec		2500 mtpa efuels		☐	☐	☑	☐	☐
Infinium				☑	☐	☑	☐	☐
Linde				☐	☐	☐	☑	☐
GTI Energy (incl. Aether fuels)	1.5 gallons/day			☐	☐	☐	☑	☐

...and many more

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WHERE CAN eREACT™ BENEFIT OUT CUSTOMERS IN THE SHORT TERM?

Archetype		Hydrogen	Methanol	Ammonia	eSAF	Waste-to-fuel	Other
Grassroot	Conventional /Grey	✓ Relevant in selected markets where NG is constrained	✗ Assumed not relevant at the moment given typical large capacity of methanol plants	✗ Assumed not relevant at the moment given typical large capacity of ammonia plants	NA	NA	Several new cases could arise as we continue to work on new applications for eREACT
	Blue	✓ eREACT™ competitive in several markets due to lower carbon capture costs (note: at scale SynCor, might still be preferable)	✗ Assumed not relevant at the moment given typical large capacity of methanol plants (SynCor more obvious choice)	✗ Assumed not relevant at the moment given typical large capacity of ammonia plants (SynCor more obvious choice)	NA	NA	
	Bio or e	✓ H2 production with biogas not common (yet). eREACT only competitive in selected markets for H2Bridge	✓ eREACT™ is capacity-relevant and offers several benefits in biogas/biomethane to bioMeOH application	✗ Assumed not relevant for eREACT at the moment (needs further investigation)	✓ eREACT™ offers the most cost-efficient eSAF (FT route) solution in the market	✓ In certain cases, eREACT can provide benefits. Value proposition still being developed	
Revamps		✓ Can benefit customers looking to reduce NG consumption, (co)process biogas or partially decarbonize (note: value proposition still under development)			NA	NA	

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HYDROGEN, AMMONIA AND REVAMPS

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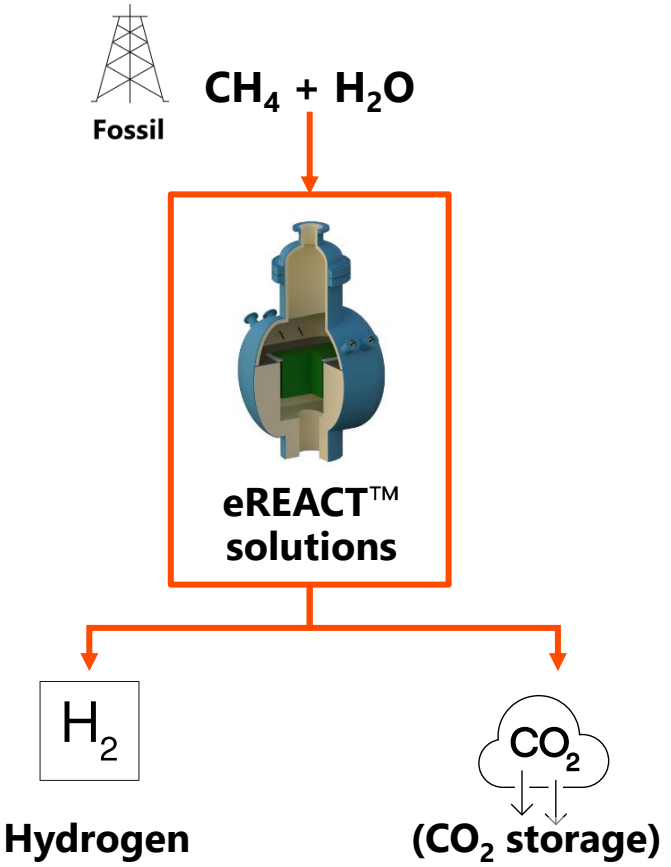
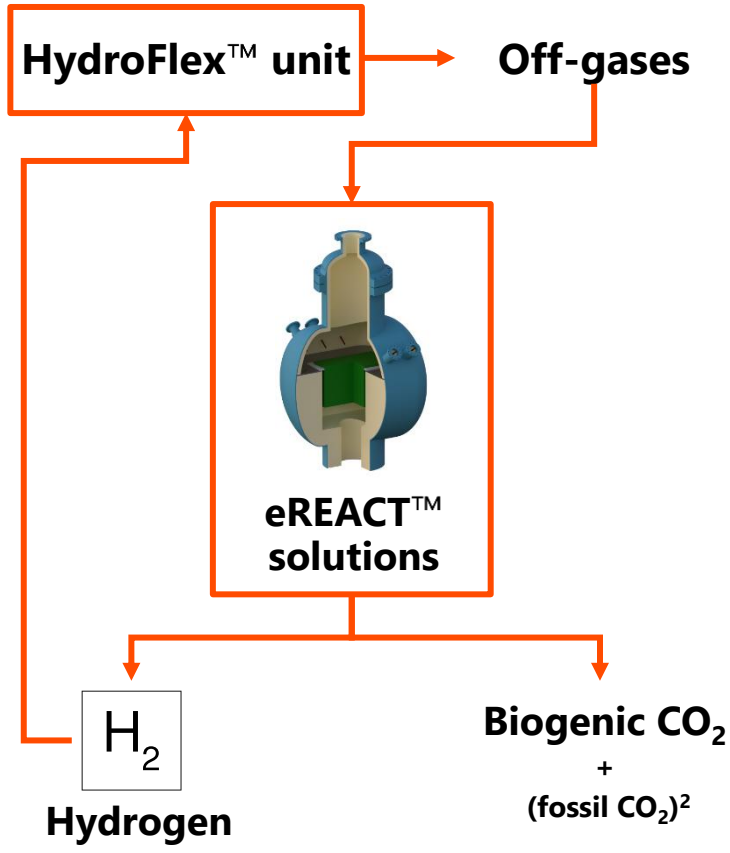
eREACT™ OPTIONS FOR HYDROGEN PRODUCTION

Blue and conventional hydrogen

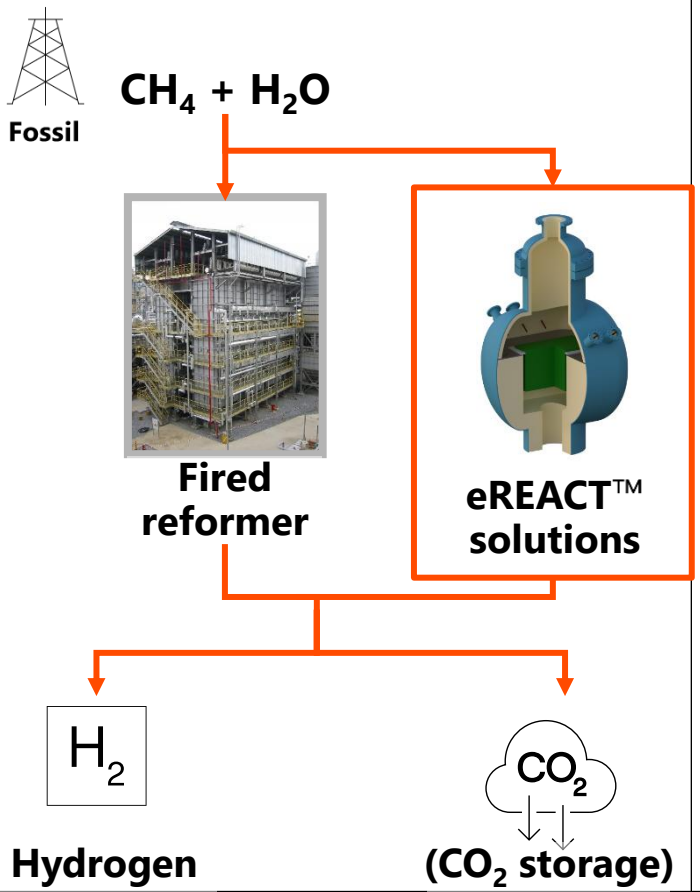


H2Bridge™[1]



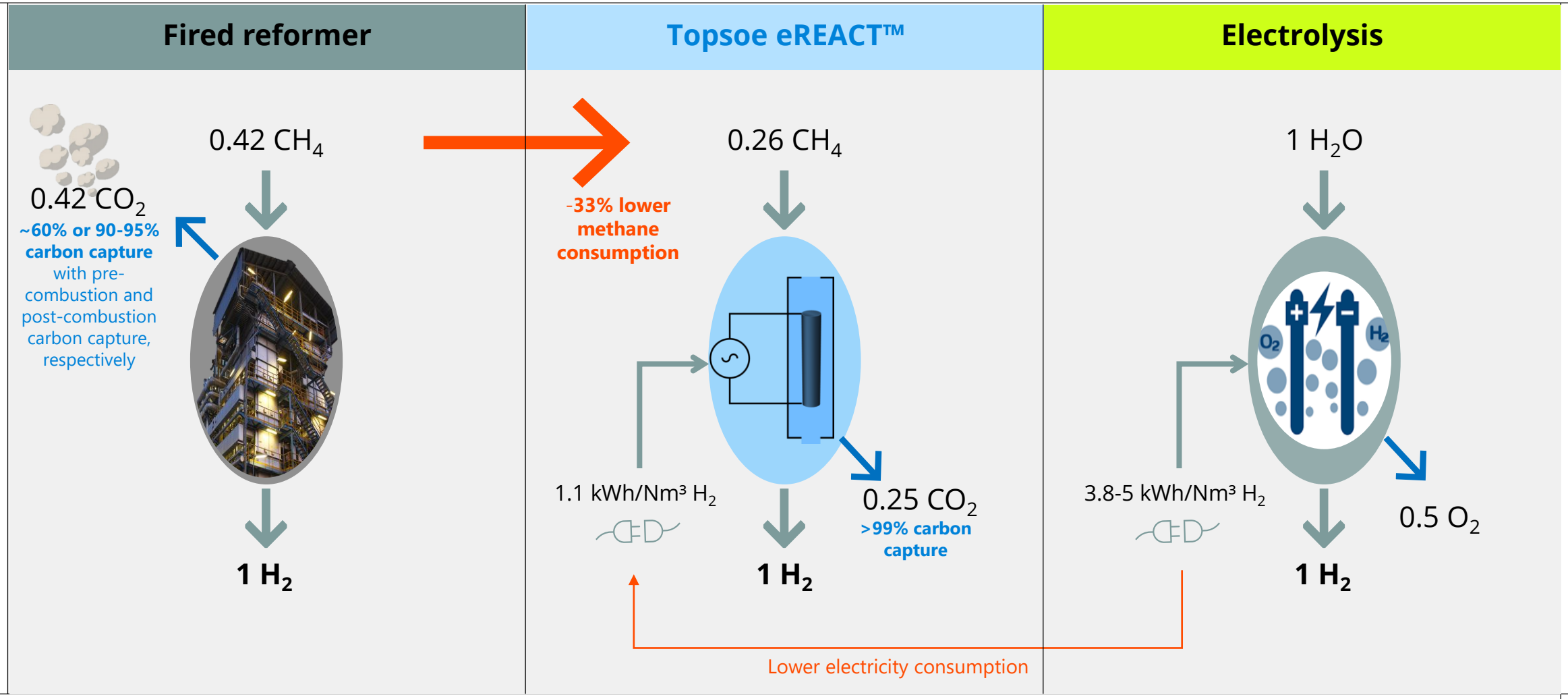
Revamps

For increase of H2 production capacity or shifting of production to eREACT™



1: eREACT can also be used to process off-gases from other processes than HydroFlex; 2: At SOR, it can be necessary to utilize natural gas to achieve the desired hydrogen yields

HYDROGEN PRODUCTION | COMPARISON OF DIFFERENT ROUTES



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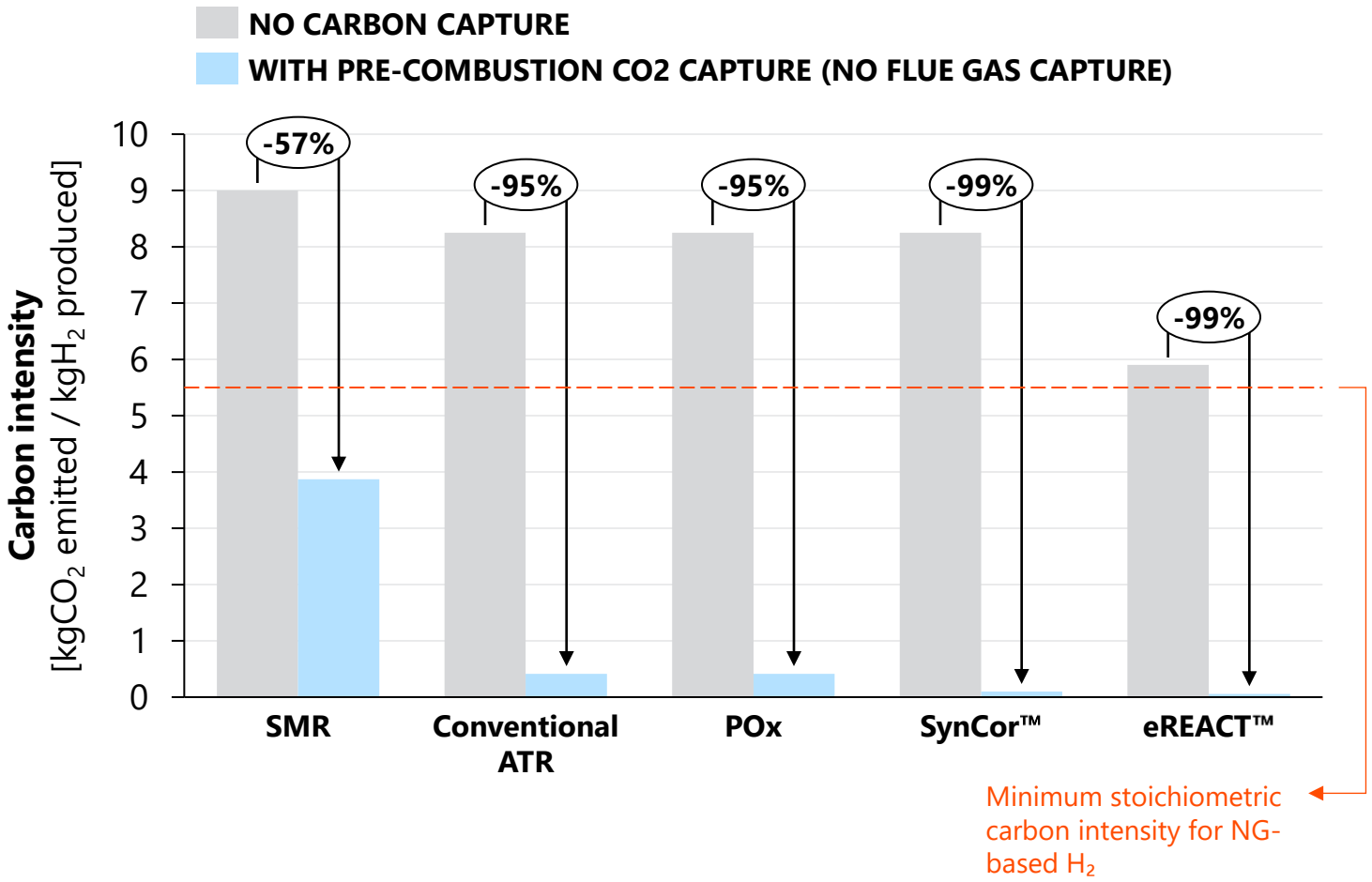
HYDROGEN PRODUCTION | eREACT™ CAN ACHIEVE 99%+ CARBON INTENSITY REDUCTION IN BLUE HYDROGEN APPLICATION



CO_2 capture is easier from the process gas or PSA off-gas rather than flue gas due to higher partial pressure

With eREACT™ blue hydrogen technology, it is possible to achieve a CI lower than **0.1 $kgCO_2/kgH_2$**

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eREACT™ REFORMING LAYOUT REDUCES COMPLEXITY - LEADING TO LESS EQUIPMENT AND EASIER OPERATION

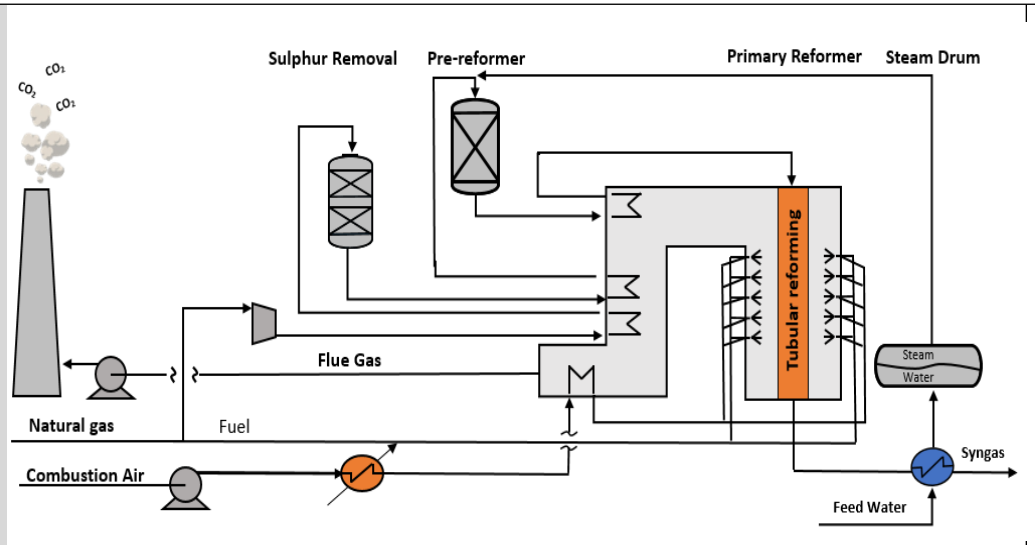
Equipment removed in eREACT™compared to SMR

Tubular reformer with:
÷ Fire Box/furnace
÷ Burners
÷ Flue gas duct
÷ Convection coils
÷ Air blower
÷ Air preheater
÷ Fuel preheater
÷ Stack

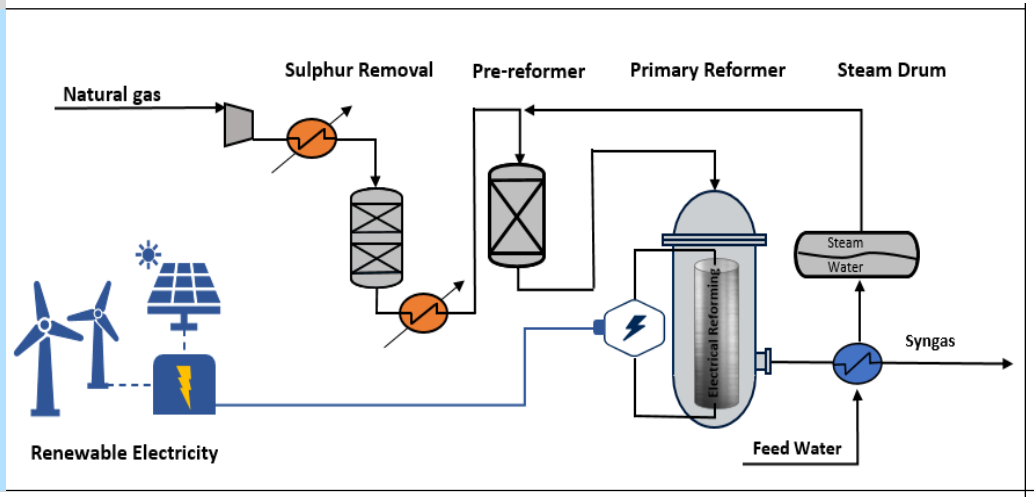
Equipment added in eREACT compared to SMR

- ✓ eREACT™ reactor (pressure vessel)
- ✓ Power supply unit
- ✓ Preheater(s)

Steam methane reforming (SMR)



eREACT™



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CASE STUDY | PARTIALLY DECARBONIZE HYDROGEN PRODUCTION THROUGH REVAMP OF REFORMER WITH eREACT™ TECHNOLOGY

With a revamp of 150 kNm^3/h hydrogen plant, **by shifting 20% steam methane reforming from tubular reformer to eREACT™ through by-pass**, you can achieve:

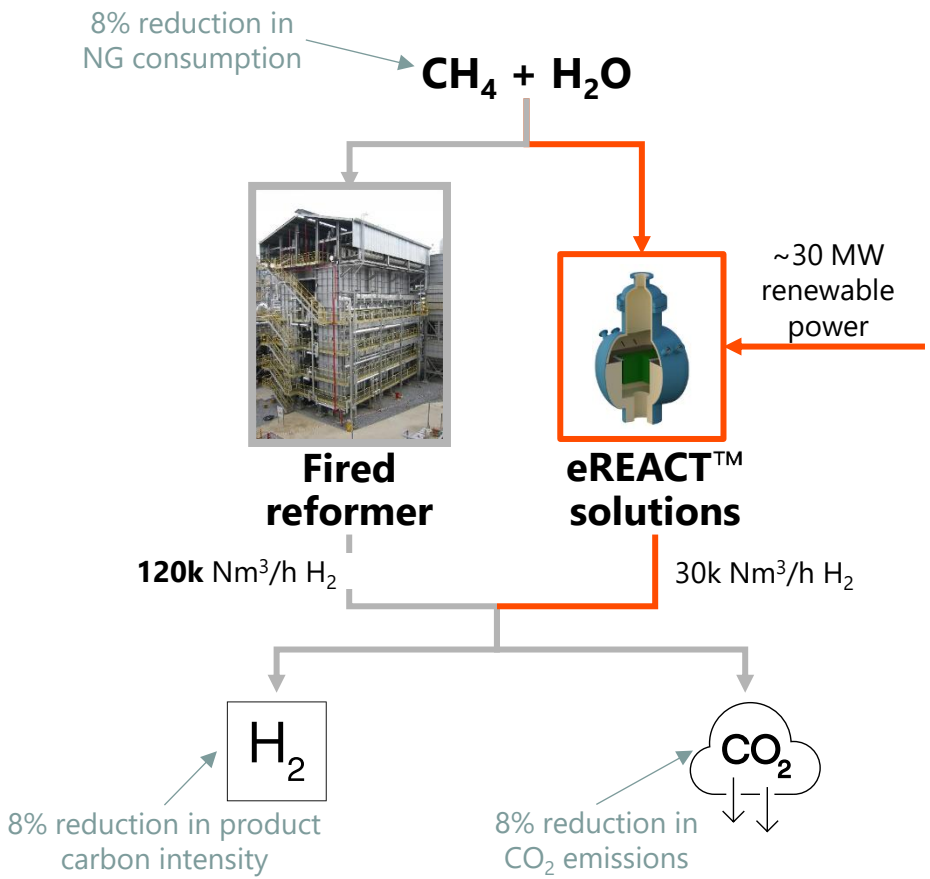
✓8% reduction in natural gas consumption

✓8% reduction in hydrogen carbon intensity

✓8% reduction in carbon emission through flue gas emissions

✓This will provide a revamp solution with no permanent changes to the existing system and few pieces of equipment to be added (eREACT™ reactor, Power Supply Unit and small boiler)

Schematic of a 150k Nm^3/h fired reformer where 20% of production is shifted to eREACT™



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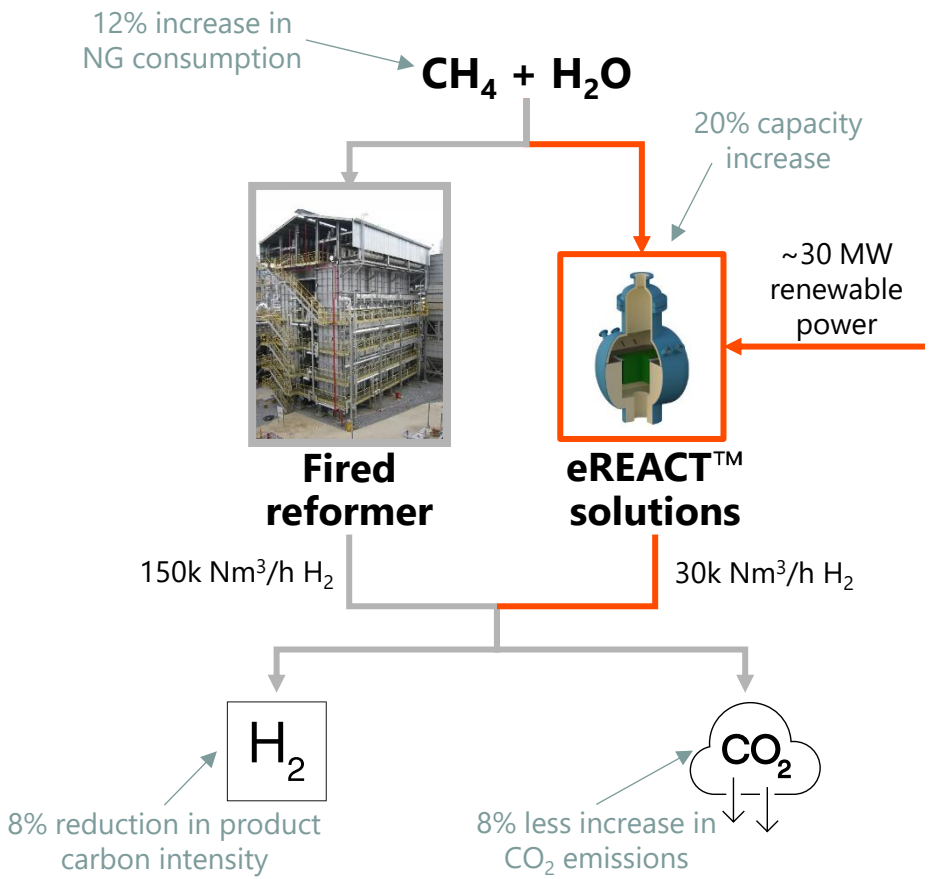
CASE STUDY | OPPORTUNITY TO INCREASE HYDROGEN PRODUCTION THROUGH REVAMP OF REFORMER WITH eREACT™ TECHNOLOGY – POTENTIAL TO LOWER CI OF HYDROGEN / AMMONIA

With a revamp of 150 kNm^3/h hydrogen plant, **for 20% additional hydrogen capacity equivalent to 30 MW renewable power eREACT™ based reforming**, you can achieve:

- ✓~30,000 Nm3/h additional low CI hydrogen product
- ✓NG consumption increase only 12% for 20% capacity expansion.
- ✓8% reduction in hydrogen carbon intensity (kg CO2/kg H2) as compared to H2 plant
- ✓8% less increase in flue gas CO2 emission for 20% capacity expansion.

Schematic of a 150k Nm^3/h fired reformer where additional 20% of production is by eREACT™



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BIOGAS/BIOMETHANE TO BIOMETHANOL

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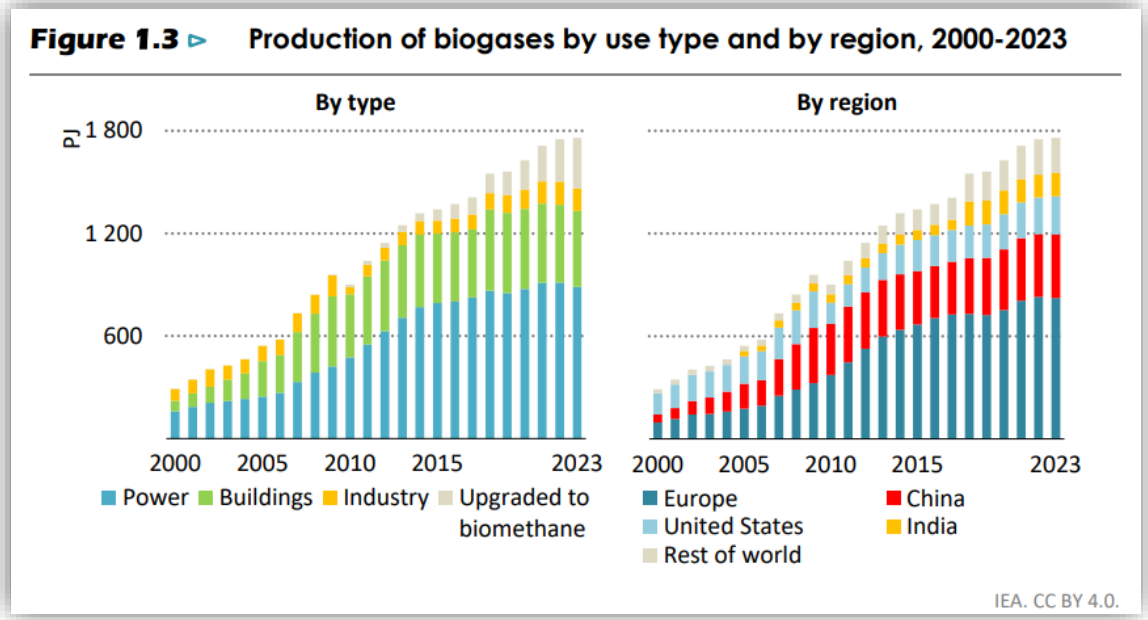
BIOMETHANOL PRODUCTION MAINLY DRIVEN BY SHIPPING AND ENABLED BY GLOBAL BIOGASES PRODUCTION EXPANSION

Biogases production

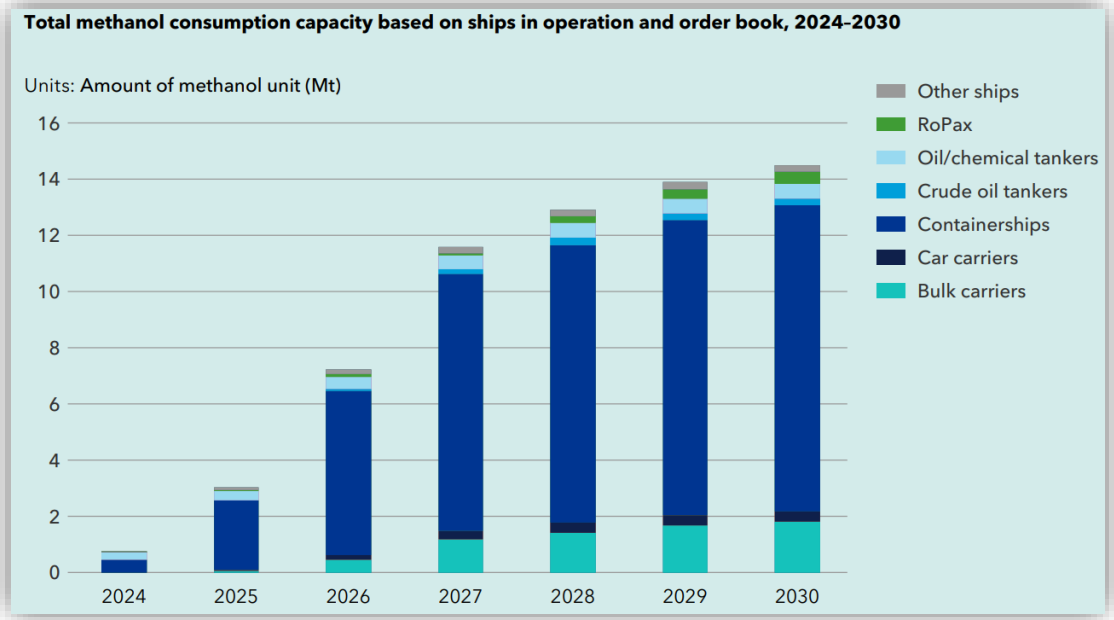
- Global production of biogases growing
- Current production is only a small fraction of total potential
- Europe produces nearly 50% of total biogases
- Increasing amount of biogas being upgraded to biomethane



Source: Outlook for Biogas and Biomethane – Analysis - IEA

Biomethanol production

- Biomethane reforming facilities account for 60% of operational renewable methanol capacity of 0.9 Mt (but only 4% of 2030 project pipeline of 45.4 Mt)[1]
- Growth of renewable (bio+e) methanol capacity depends on regulatory developments in maritime, chemicals, aviation and road fuels

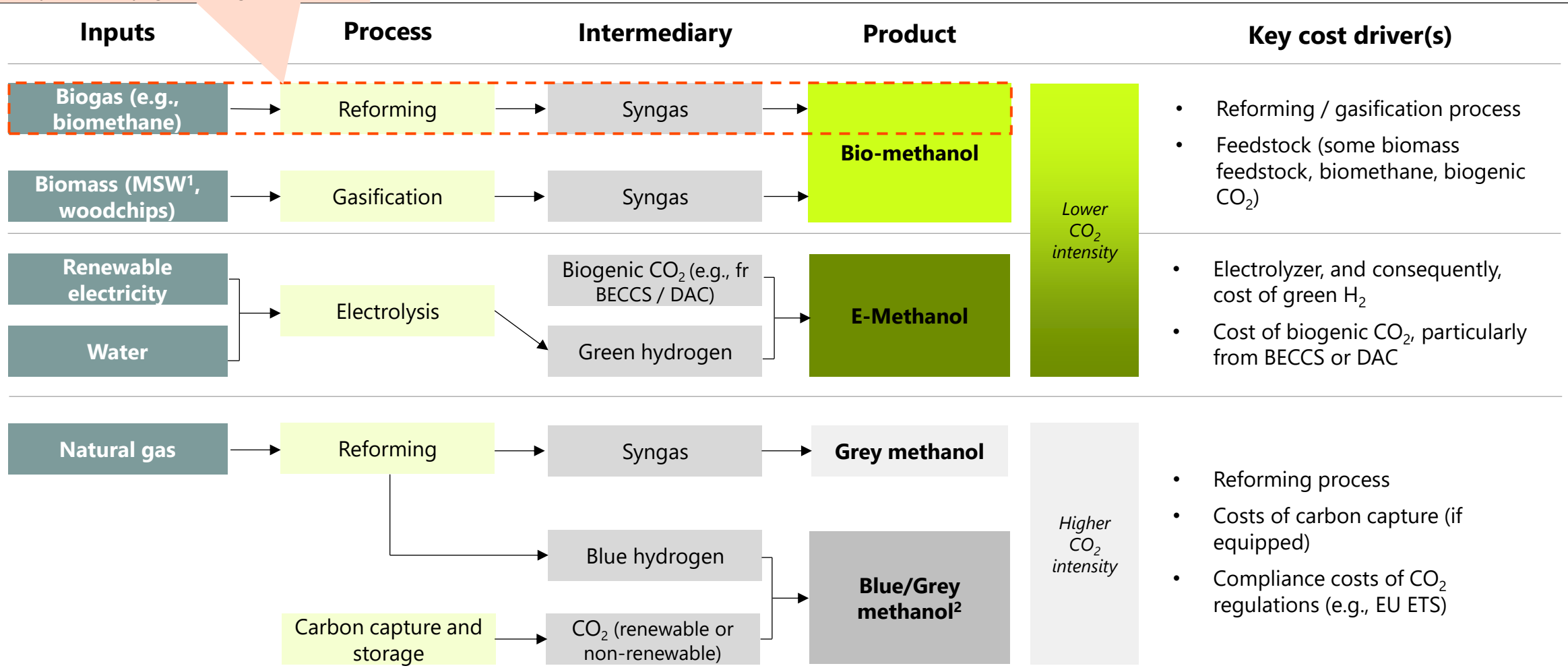


Source: Methanol fuel in shipping

1: Source: GENA Solutions Oy. Note BioMeOH can also be produced from gasification.

METHANOL CAN BE PRODUCED THROUGH ROUTES – EACH WITH ITS OWN COST DRIVERS

Focus of this section. However; note that eREACT™ can also be applied in some cases when you have syngas from gasification



Notes: 1. Municipal Solid Waste **2.** IRENA classifies methanol from blue hydrogen and CO_2 from CCUS as "blue methanol", however it is uncertain if such production route will give lower carbon intensities from an LCA calculation perspective, particularly if the captured carbon is from non-biogenic sources (e.g., natural gas power production and potentially steel / cement manufacturing)

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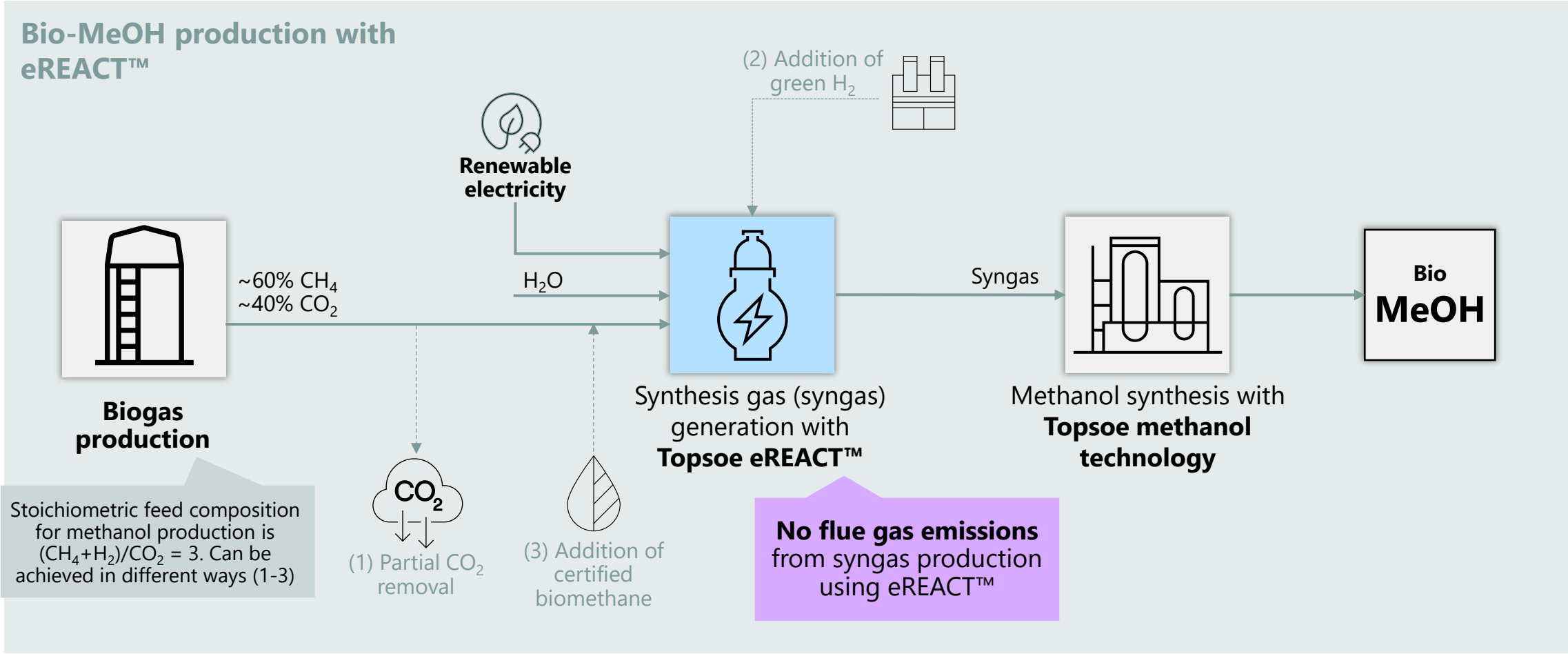
BIO-METHANOL FROM BIOGAS | eREACT™ REMOVES FLUE GAS EMISSIONS FROM SYNGAS PRODUCTION AND MAXIMIZES FEED EFFICIENCY



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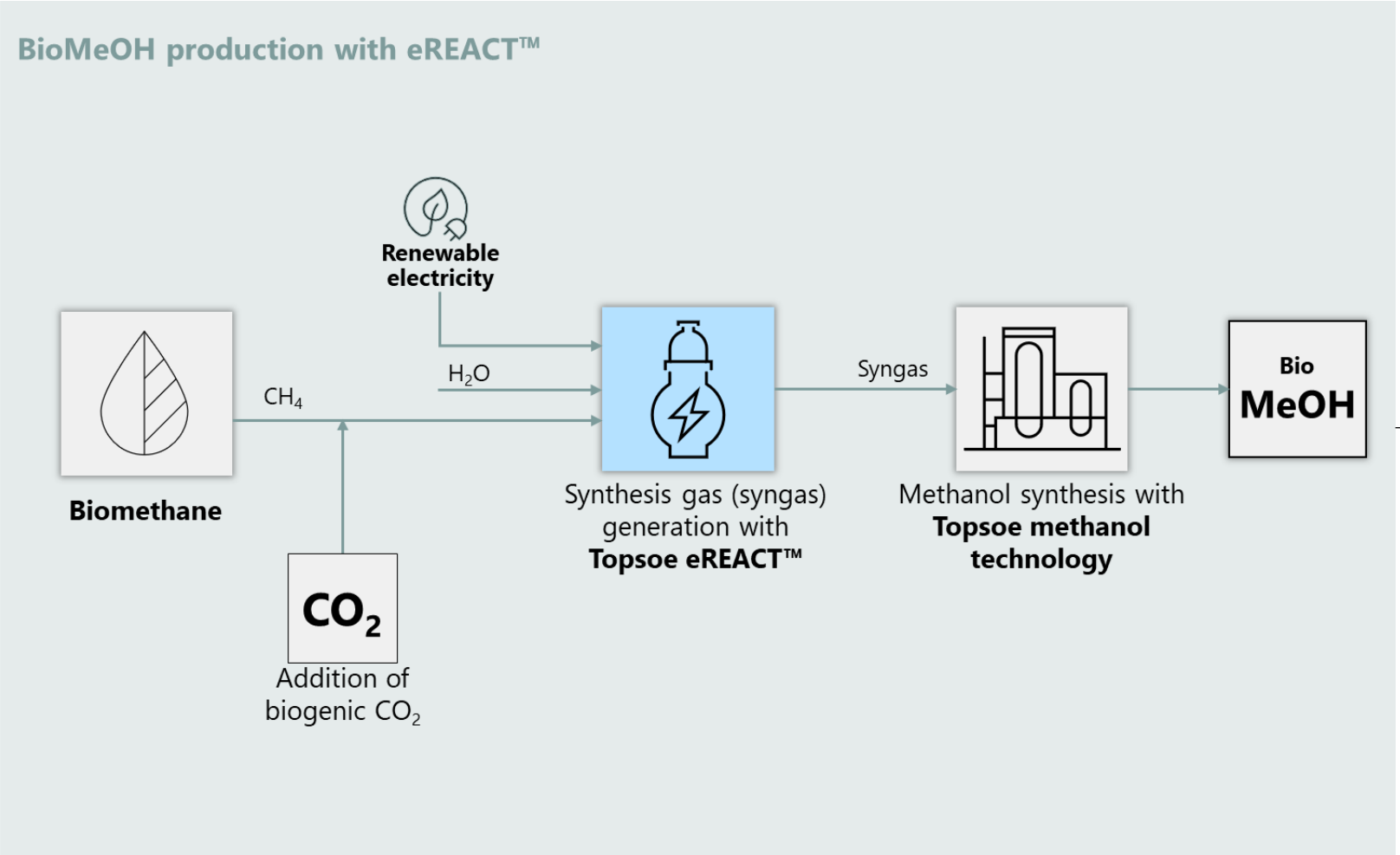
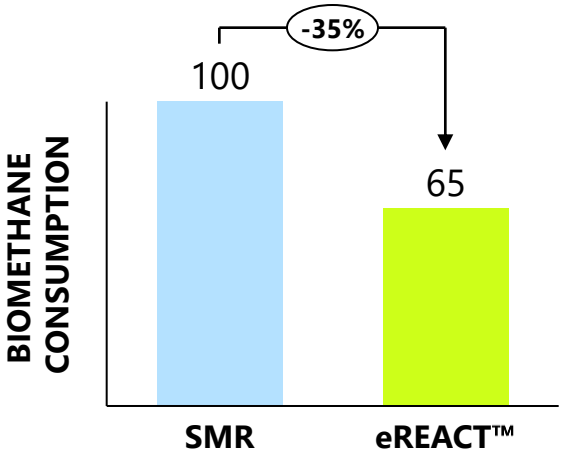
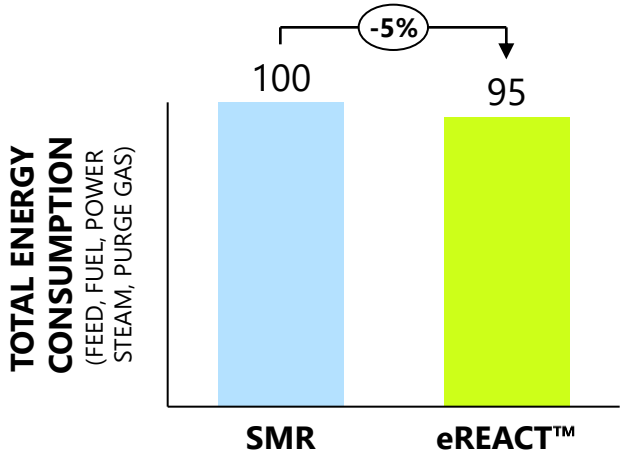
BIOMETHANOL FROM BIOMETHANE | eREACT™ REMOVES FLUE GAS EMISSIONS FROM SYNGAS PRODUCTION AND MAXIMIZES FEED AND ENERGY EFFICIENCY







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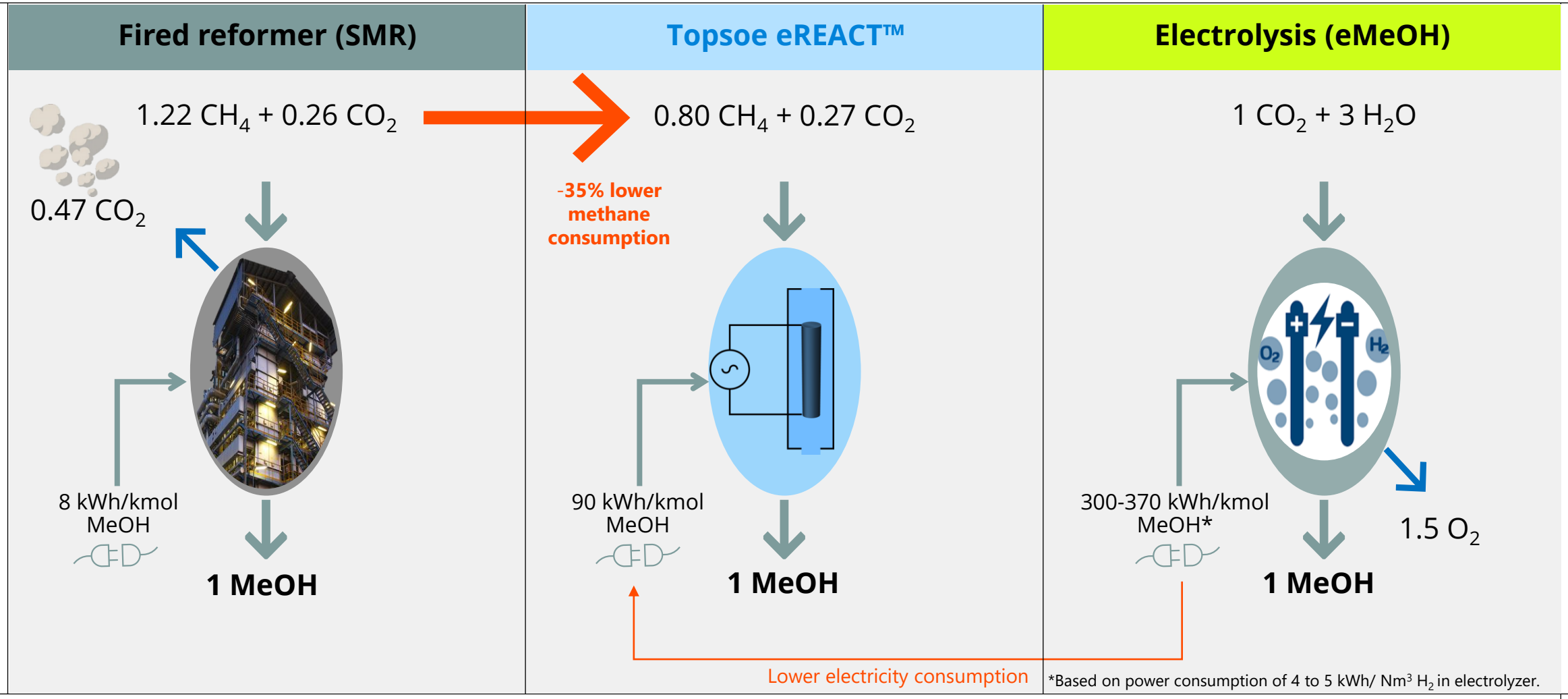
METHANOL PRODUCTION | COMPARISON OF DIFFERENT ROUTES



*Based on power consumption of 4 to 5 kWh/ Nm^3 H_2 in electrolyzer.

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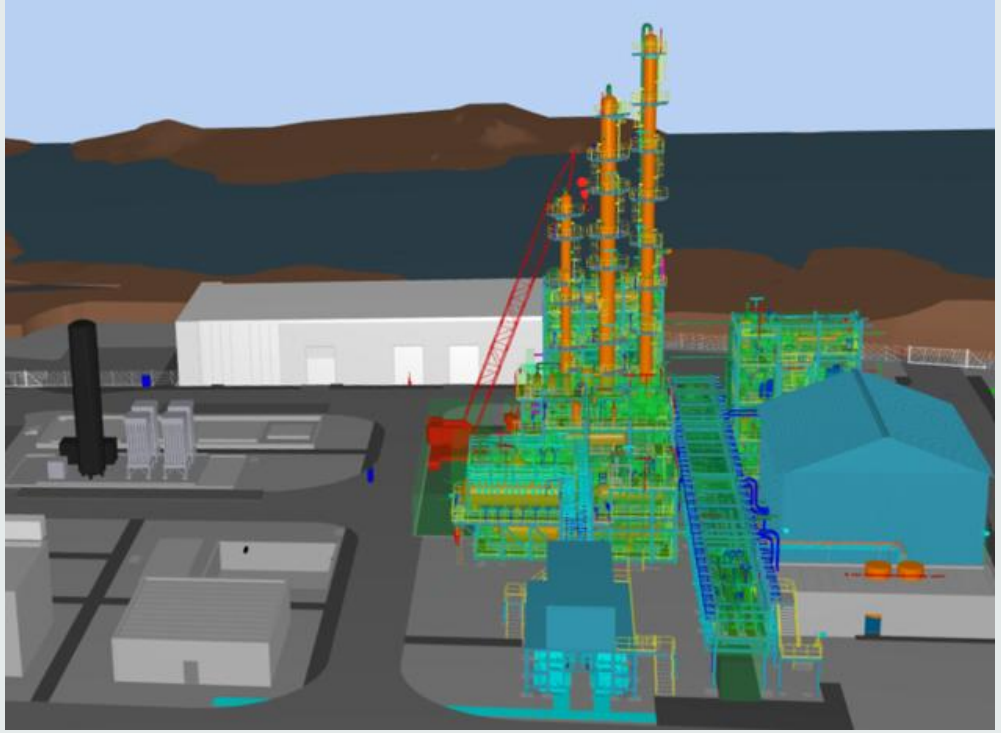
NEXTFUEL BIO-METHANOL PLANT PROJECT OBJECTIVES, STATUS AND MILESTONES

Nextfuel

Project objectives

- **Demonstrate first full scale eREACT™ unit**
- **Demonstrate all aspects of novel technology in conjunction with downstream methanol synthesis,** i.e. from start-up through normal operation to shut-down and maintenance & service
- **Prove commercial viability of eREACT™-based methanol production**



Key milestones of Nextfuel project

2024

Nov ’24: **BEDP completed** by TOPOSE

2025

May ’25: **FEED completed** by TOPOSE and modularization subcontractor WASCO

Jun ’25: **Topsoe submits firm bid proposal**

2026

May ’26: **All relevant site permits expected**

Q2 ’26: **FID expected**

Q4 ’26: **FC expected**

2028

Q2 ’28: **ModuLite™ modules on site**

Q4 ’28: **Commissioning and start of operation planned**

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eSAF AND WASTE-TO-FUELS

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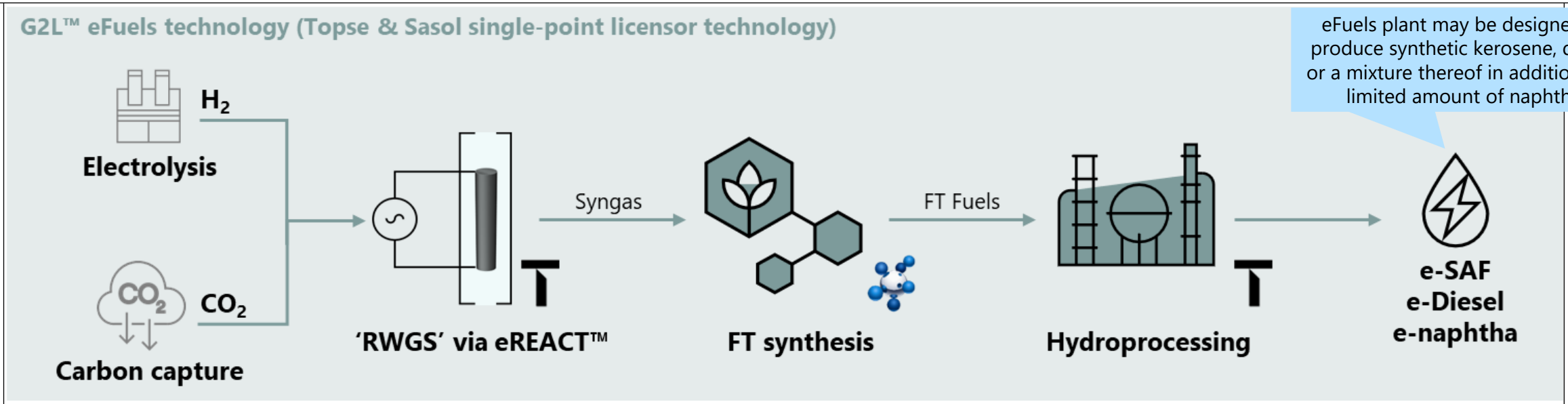
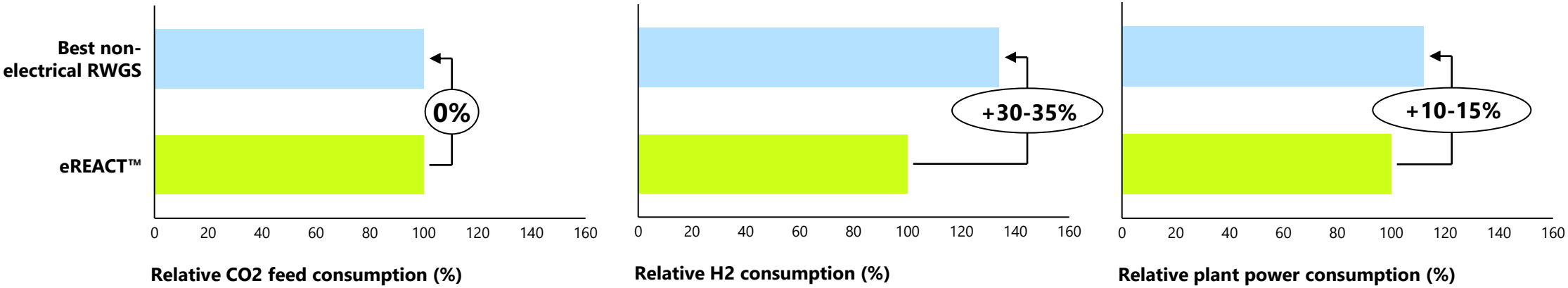
eFUELS PRODUCTION | LOW HYDROGEN CONSUMPTION IN eREACT™ TECHNOLOGY IS KEY TO MAKING IT THE MOST EFFICIENT TECHNOLOGY FOR eFUELS PRODUCTION





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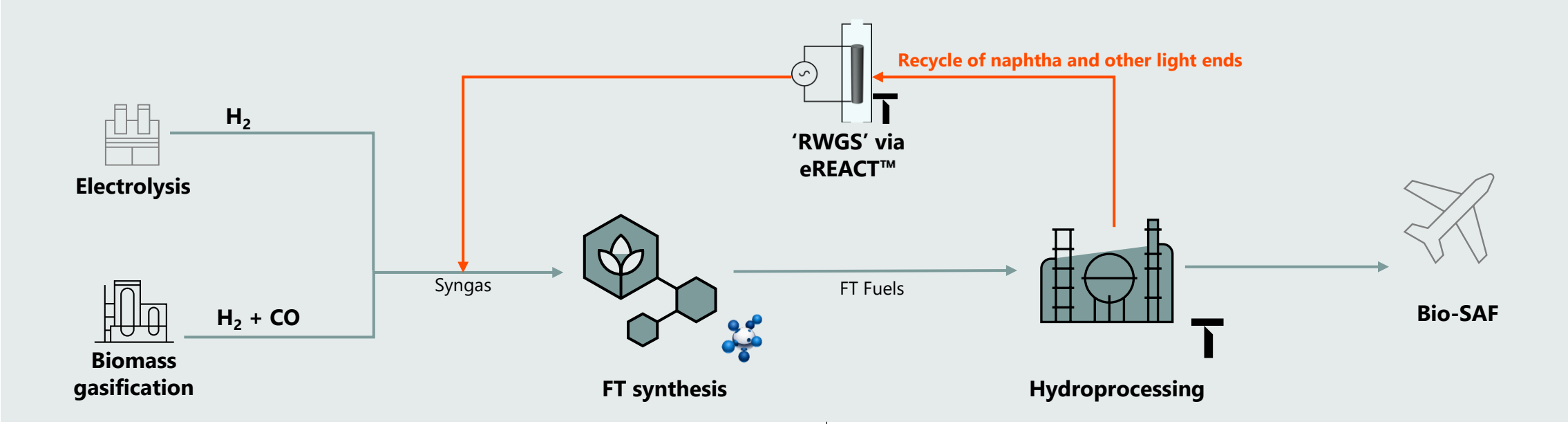
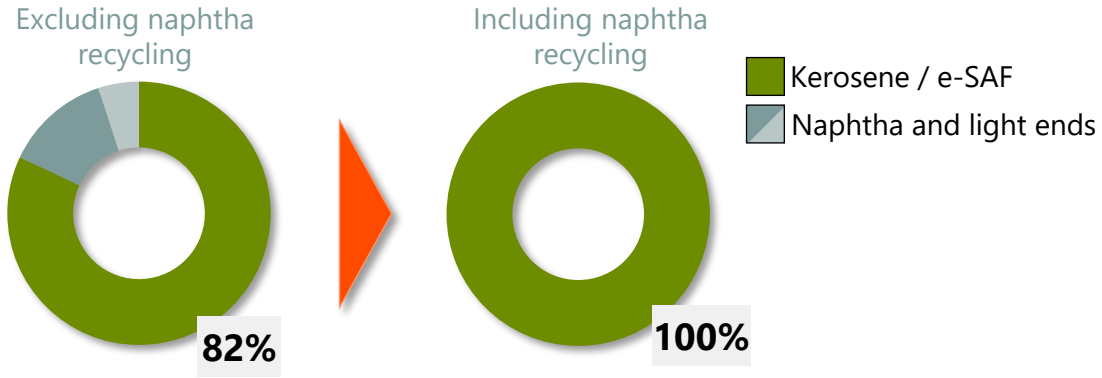
WASTE-TO-FUELS | YOU CAN MAXIMIZE SAF YIELD AND REDUCE OVERALL HYDROGEN AND ELECTRICITY DEMAND WITH G2L™ BIO-FUELS SOLUTION BY RECYCLING LIGHT ENDS AND NAPHTHA





For maximum carbon, hydrogen and product efficiency

- >95% carbon efficiency and 25% higher SAF yield
- 97% theoretical hydrogen into sellable product
- 100% production of ASTM certified synthetic kerosene

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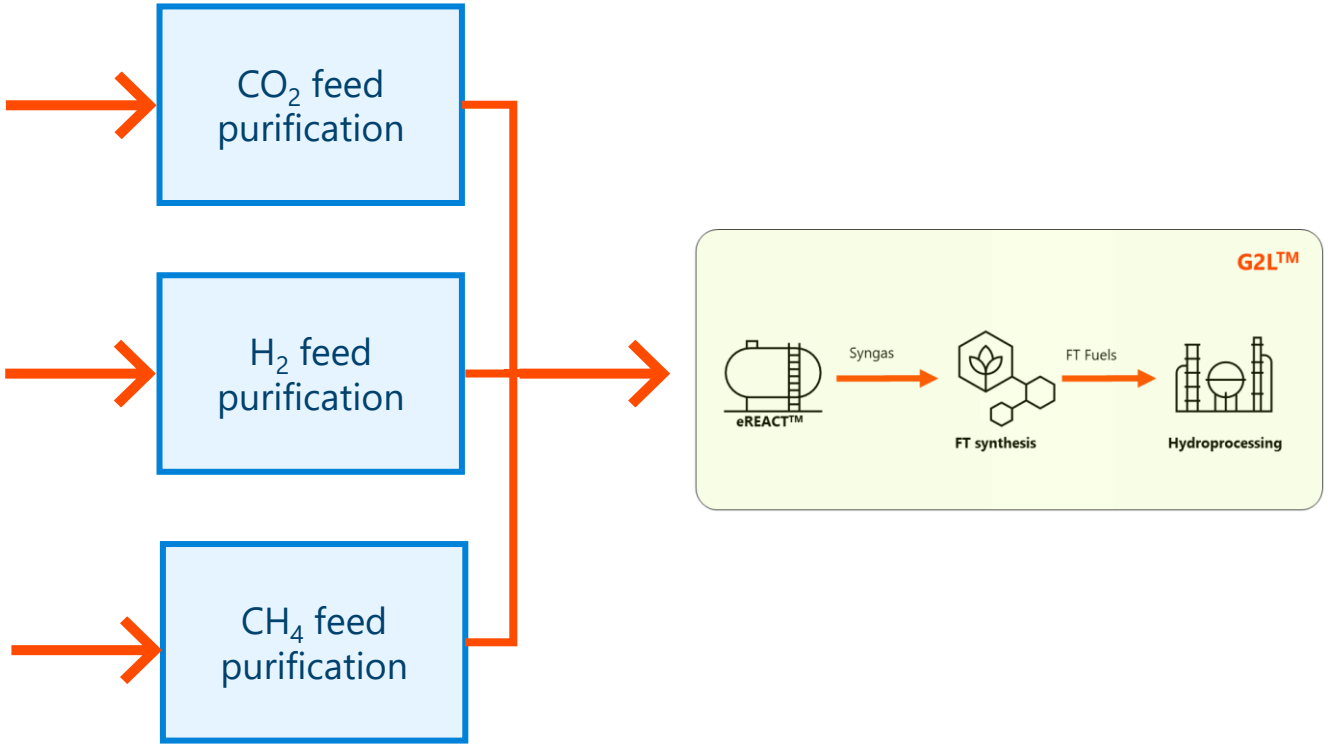
WE ARE SEEING A GROWING INTEREST IN MIXED FEED WITH BIOGAS/BIOMETHANE AS C AND H CARRIER

Growing interest in mixed feed layouts with use of biomethane as a carbon and hydrogen carrier to **add flexibility and balance high electricity price periods**

Based on knowledge from eFuels and NG purification. Each feed-stream will be purified separately

Based on same control principles as in Foulum the **oxygen/carbon and hydrogen/carbon ratios at eREACT™ inlet can be maintained ensuring constant syngas quality** (see our newly published paper: Link)



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eREACT™ PILOT PLANT OPERATION DEMONSTRATES INTERCHANGEABLE OPERATION BETWEEN BIOFUELS AND eFUELS PRODUCTION

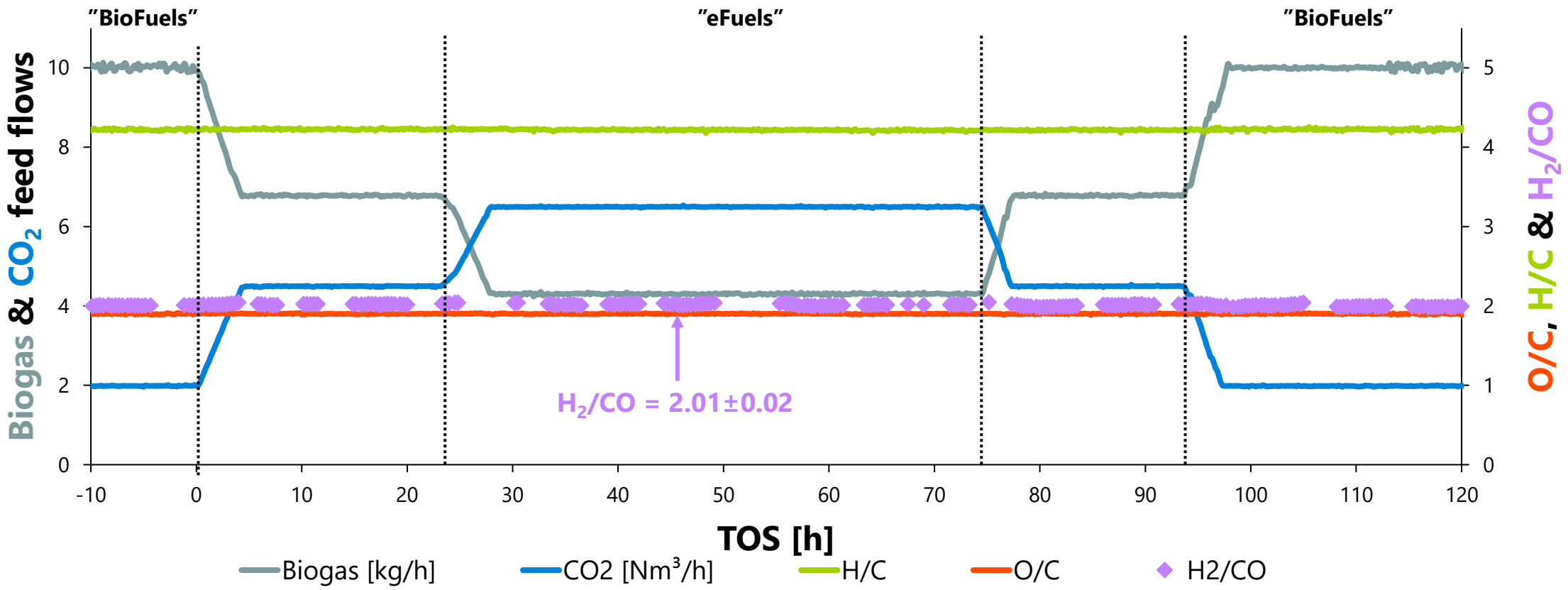

Figure showing data from the eREACT™ pilot plant with a transition from eSMR (BioFuels) to eRWGS mode (eFuels) online through the change of feed gas composition

Internal

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OVERVIEW OF TPP DEMONSTRATION PLANT OBJECTIVES, STATUS AND MILESTONES

Project objectives

- Bring **fully integrated production of synthetic kerosene** (SAF for test flights) on the Topsoe and Sasol technology G2L™ platform
- **Demonstrate all aspects of the technologies** from start-up through normal operation to shut-down and maintenance
- **Demonstrate scale-up of all process steps**
- Demonstrate wax production with high alfa-value on latest generation FT catalyst from Sasol

Picture of the on-going ground preparation at Leuna chemical park during 2025

Key milestones of TPP project

2024

Oct ’24: **Joint basic engineering** by TOPOSE/Sasol SPL and Griesemann initiated

2025

Jan ’25: **Detailed engineering and procurement** of proprietary hardware and catalyst

Jan ’25 – **Detailed design and fabrication** of the eREACT™ equipment

May ’25: **PIDs and HAZOP completed**

Q2-Q3 ’25: **FID, site preparation and construction start**

2026

2027

Q4 ’27: **Commissioning and start of operation**

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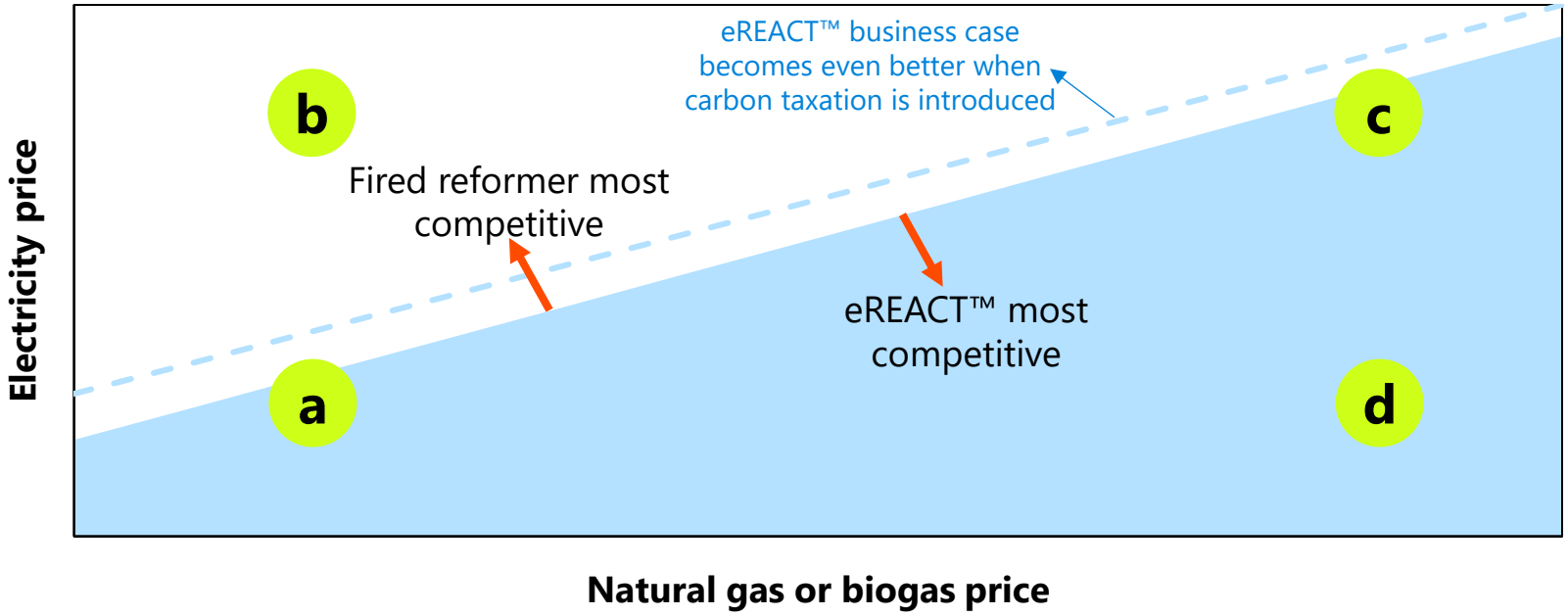
WITH THE RIGHT MIX OF POWER AND GAS PRICES, eREACT™ GIVES LOWER LEVELIZED COST OF PRODUCT ALSO IN THE ABSENCE OF CARBON TAXATION – THE STUDY FOR HYDROGEN



a & **c**

eREACT™ and fired reformer costs comparable. Other value drivers should be considered to make best decision. Chose eREACT™ for lower carbon intensity, higher feedstock flexibility and easier O&M

b

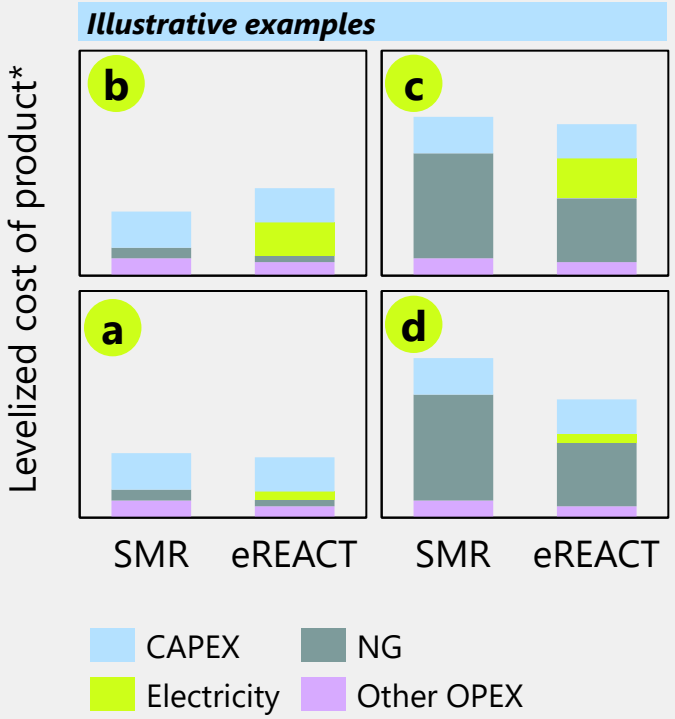
Fired reformer most cost competitive option based on levelized cost of product. Chose eREACT™ if cost is not main value driver and additional eREACT™ benefits outweigh costs

d

eREACT™ most cost competitive option based on levelized cost of product. eREACT™ comes with additional benefits such as lower carbon intensity, higher feedstock flexibility and easier O&M

Levelized cost of product depends to a high degree on gas price

In the right markets, electrification of heating can save substantial costs



* Examples of products are hydrogen, ammonia or methanol

Internal

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WHICH CUSTOMERS ARE WE LOOKING TO TALK TO?

Target customers

Feed gas constraints (volume or price)

- Relevant for majority of biogases-relevant markets
- Especially relevant for NG importing markets

Access to low-cost (renewable) electricity

- Relevant for markets with large renewable build-out
- Especially relevant for markets where electrolysis remains expensive

Incentive for carbon emissions reduction

- Relevant for markets and applications with carbon reduction regulation

Priority in short term given to balance sheet customers

Target projects

Can be grassroot or revamps

Can be small (>3MW) or larger (up to 50 MW)

We might need to guide customers more on how project can look

Please help us refine our go-to-market and develop our standard eREACT™ offerings by providing us with feedback from customers on the following:

- Main pain points (current and future)
- Topics occupy their company (energy independence, decarbonization, etc.)
- Future plans for expansion, closures, diversification etc.
- Feedback on eREACT™ technology (positive and negative)
- And anything you find relevant

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SLIDE DUMP OF UNFINALIZED WORK THAT CAN BE USED FOR INSPIRATION

TOPSOE

Internal

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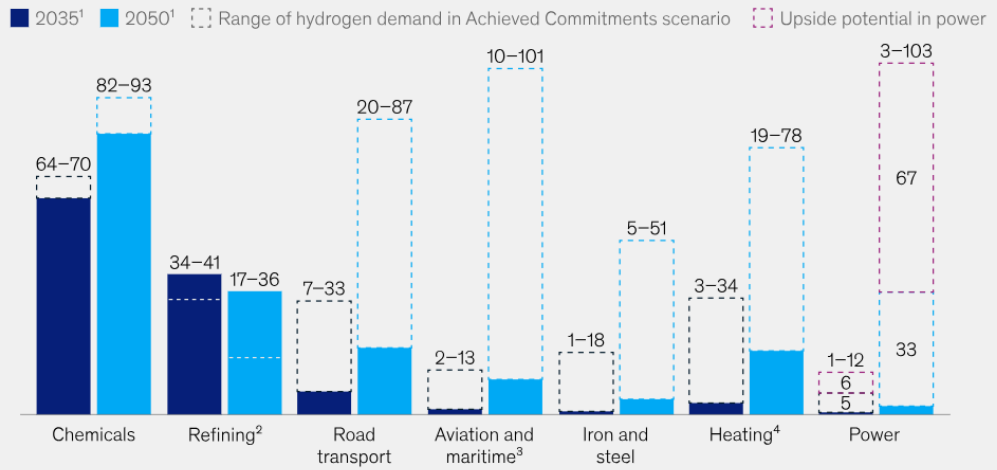
CHEMICALS & REFINING CONTINUE TO PROVIDE LARGE PART OF ALL H2 DEMAND - ALSO IN 2050

WHY WE ARE FOCUSING TODAY ON CHEMICALS AND REFINERY FOR DOWNSTREAM USE

There is a large range of uncertainty in the projected uptake of hydrogen in emerging sectors.

Global hydrogen demand by sector, range between Fading Momentum to Achieved Commitments scenarios, Mt per year of hydrogen equivalent



[1]Fading Momentum scenario.
[2]Includes conventional fuels refining and biofuels hydrogenation and refining.
[3]Aviation and maritime include direct use of hydrogen and hydrogen-derived synfuels including kerosene, diesel, methanol, gasoline, and ammonia. The category also includes some hydrogen-derived synfuels in road transport.
[4]Includes hydrogen demand for heating in other industry and buildings.
Source: McKinsey Energy Solutions' Global Energy Perspective 2023

McKinsey & Company

End-use of hydrogen	**Total global demand 2023 (Mt)**	**Potential Topsoe customers** (broadly in H2, not necessarily just eREACT or blue H2)
Chemicals production (NH3 & MeOH)	46	Chemicals producers, project developers, merchant H2 producers
Refining (traditional)	43	Refiners, merchant H2 producers
Steel manufacturing	5	Steel manufacturers, merchant H2 producers
Biorefining	<1	Refiners, bio-refiners, project developers and merchant H2 producers
eFuels	-	Project developers and merchant H2 producers
Long-distance transport (direct use)	<1	Project developers and merchant H2 producers
Electricity storage and generation	<1	Project developers and merchant H2 producers
High-temperature industrial heating	-	Industrial players?

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Please review input AND add missing hypotheses/questions

CUSTOMER ARCHETYPE 1 – HYDROGEN
JOBS TO BE DONE & CHARACTERISTICS

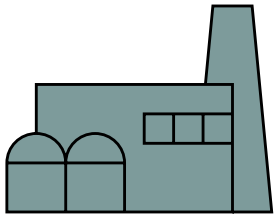
Jobs to be done

Chemicals:

- Operate reformers stably and continuously to produce hydrogen for NH3 and MeOH production (i.e. high uptimes and low fluctuations in capacity)
- Primary focus is on operating at lowest possible LCOH, since carbon is imbedded in the product and typically not incentivized/mandated to reduce product carbon emissions (unless value of low-carbon product can make up for increased price)

Refinery (incl. biorefineries):

- Operate (typically several) H2 units to match the down-stream needs. Typically, this means operating with fluctuations in unit loads (even down to 30%)
- Likely to investigate revamp options to increase capacity or shift load to decarbonize since existing unit sizes can make it a feasible option. Might also consider purchase of merchant hydrogen through pipelines to cover need
- Comply with and/or minimize cost impact of carbon-reduction legislation in relevant region



Producers of hydrogen for own down-stream on-site applications (traditional Topsoe customers)

Refiners, ammonia and methanol producers

Characterized by

Chemicals:

- Large front-end reformer capacities (~200kNm3/hr H2/2000 MTPD NH3) - meaning that also small %-wise expansions of capacity can be large in absolute numbers
- Not driven by carbon reductions targets unless it increases value of product
- High risk-aversion

Refinery (incl. biorefineries):

- Small(er) H2 unit capacities (average 100k Nm3/hr, but 20-50k Nm3/hr also common)
- High(er) risk-willingness (compared to chemicals)
- Pressure by legislation to decarbonize

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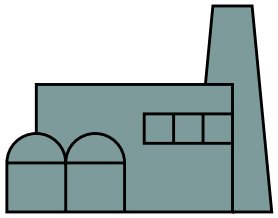
Please review input AND add missing hypotheses/questions

CUSTOMER ARCHETYPE 1 – – HYDROGEN
PAINS AND GAINS

Customer pains

- Pressure to produce hydrogen at lowest price while simultaneously lowering carbon intensity is difficult given that LCOH using grey solutions remains lower
- Low availability of RE in grid and through PPAs can reduce the financial upside from investing in a greener production (depending on region)
- Feedstock availability (e.g. NG in Europe) creating cost uncertainties in existing operations and posing risks in bankability of future investments
- Low risk-willingness in industry making it difficult to invest in low TRL solutions (as well increase availability of more mature solutions) that could solve some of the other pains
- Lack of alignment around environmental legislation in various regions creating uncertainty around business case of green production
- To be investigated: is the ramp (up and down) load and speed a pain point? For which customers and why?



Producers of hydrogen for own down-stream on-site applications (traditional Topsoe customers)

Refiners, ammonia and methanol producers

Customer gains

- If renewable is easily/cheaply available in customers' region, electrifying (part) of the operations could provide cost benefits
- If customer has low-value off-gas stream available, eREACT could help turn that into a higher value low-carbon hydrogen

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CUSTOMER PAINS AND GAINS: BIOGAS TO BIOMETHANOL
FROM CUSTOMER SIDE

Customer Pains

Difficult to secure off-takers and funding for BioMeOH projects

- Without offtake agreements, it is not possible to finance the projects
- Willingness to pay from potential off-takers is low due to lack of political incentives and due to fierce competition from players in especially China that are offering bioMeOH at low prices (down to 450 EUR/t)
- Only projects with low levelized cost of product (LCOP) can secure offtake while also showing a positive business – both a pre-requisite to secure funding and reach FID
- At large product volumes, it is easier to achieve a lower LCOP through economy of scale; however, it is difficult to achieve large CAPEX investments in a time of large economic and political insecurity - smaller projects might therefore be more likely to receive investment (if LCOP can simultaneously be reduced)

The most future-proof project (from feed to product) is not easy to identify (and maintain) in the current global geopolitical and economic landscape

- There is currently large legislative uncertainty and geographic differences in energy policy and strategy, making it difficult to predict the future landscape and which final products are a good economic bet for project developers
- There is currently large uncertainty around whether there will be a large adoption of BioMeOH as shipping fuel both due to legislation (IMO net-zero shipping being postponed) and competing pathways for (low-carbon) shipping emerging (e.g. LNG and LBG)
- Biogas markets are still nascent in many areas making it difficult to predict how the feedstock landscape will look in the future (size of plants, gas gird expansion etc.). This has large implications on location and design of project

Topsoe's eREACT's pain reliefs

Feed and energy efficient solution for….

- Minimize use of valuable biogas/biomethane feed through electrification of heating to achieve lower LCOP compared to conventional reforming (30-40% less biomethane consumption, 50-55% higher methanol production and X% higher overall energy efficiency)
- More efficient/smart use of high value feed to achieve target product specifications (needs testing) – also maybe something about the inherently lower CI
- Modular and scalable solution that can be designed at desired capacity (from 3 MW/50 TPD to 100MW/1200 TPD BioMeOH)
- High feedstock versatility to enable the best fit for the project economics
- Highly experienced Topsoe team that can support in exploring and assessing the best options to bring down the costs of the project
- Topsoe team is experienced in assisting developers with their financiers to alleviate any concerns about technology risk. Having Topsoe on-board as a licensor can in itself (due to Topsoe's experience and reputation) be helpful for the project

Your partner along the journey from idea to operation

- Topsoe team can assisting in weighing options for contingency plans both in terms of scale and technology
- Topsoe can assist in connecting with value chain players
- Topsoe's expertise can help you all the from deciding on the project layout to ensuring that the unit operates as intended

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